

Calibrating Discrete Entropy Prediction Floors for Urban Crime Counts: A Multicity Evaluation

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Abstract

Urban forecasting systems increasingly draw on mobility and other dynamic data, yet improved test accuracy alone cannot show whether an added source lowers an information-constrained error bound or merely benefits a particular model. We develop a finite-sample evaluation framework based on the Discrete Entropy Lower Bound (DELB) for integer-valued prediction. DELB maps conditional Shannon entropy to an MSE lower bound through the integer-lattice maximum-entropy envelope. The empirical analysis separately examines sparse-state estimation bias, evidence from prespecified randomization schemes, and held-out performance. We apply the framework to 503,128 crime events and 86,269,976 flow-eligible bicycle trips from Washington, DC, New York City, and Vancouver during 2020–2022, aggregated into 1,241,768 1 km grid–day observations. In the primary high-dimensional specification, the observed conditional mutual information did not exceed the temporal, spatial, circular-shift, or baseline-state-matched reference distributions. The positive estimated DELB reductions are therefore finite-sample point estimates, not calibrated evidence of source-specific information: they were small over the complete crime-supported populations and 7.9%, 17.3%, and 18.0% within bicycle-covered areas. A lower-cardinality sensitivity detected a small calibrated difference only in New York City (0.00105 bits above the matched-reference mean; BH-adjusted $q = 0.003$), showing that empirical resolution depends on state complexity. In the scale–sample reference experiment, semi-synthetic null rejection was near nominal, whereas sparse-state settings were conservative (Type I error 0.011–0.014) and power was limited for small effects; held-out gains varied across cities and models. The results support a staged evaluation of new urban data sources: the estimated bound change, randomization-based evidence, and realized forecasting gain should be reported separately.

Keywords: urban crime prediction; urban mobility; smart-city analytics; shared bicycle data; discrete entropy lower bound; conditional mutual information; finite-sample calibration; spatiotemporal data mining

Received:

Revised:

Accepted:

Published:

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1. Introduction

1.1. The Data-Value Problem in Urban Crime Count Forecasting

Urban crime forecasting increasingly combines historical incidents with mobility traces, calendar variables, environmental measurements, and other dynamic urban data. These sources can provide richer representations of activity around a location, but they also require acquisition, cleaning, spatial alignment, maintenance, and governance. For

a city agency or an analyst, the relevant question is not simply whether a fitted model achieves a lower test error after a new source is added, but whether that source reduces uncertainty about future crime counts beyond information already available from location, crime history, and the calendar.

These two questions are not equivalent. A change in held-out error depends on the fitted model, hyperparameter search, training sample, target definition, and validation period. It may reflect useful new information, a better match between the added variables and one model class, or sampling variation. In contrast, a model-independent prediction benchmark is determined by the joint distribution of the target and the information available when a forecast is issued. Comparing achieved error with such a benchmark can help distinguish information-constrained uncertainty under the specified target and information set from error that may remain reducible through improved modeling.

Crime count forecasting makes this issue practically important. Incident counts are nonnegative integers, are often zero at fine spatial and temporal resolutions, and are evaluated under losses such as mean squared error (MSE). The forecasting literature has progressed from models using urban metrics [4] to representations of temporal dependence [6] and multiscale spatial structure [8]. Reviews of spatial forecasting [1], multiscale designs [2], and machine- and deep-learning methods [3] show how widely targets and evaluation settings vary. Their comparative results are essential for model selection, but they do not identify how much error is unavoidable under a specified information set. This limitation becomes more consequential when an operationally demanding new urban data source appears to improve one or more fitted models.

We study this problem using shared-bicycle activity as a candidate source of additional information. The motivation is observational rather than mechanistic: station departures and arrivals provide high-frequency, spatially resolved traces of one form of local street activity and exposure. They may therefore help describe when and where potential targets, guardians, or other users of public space are present beyond what recent crime and the calendar already reveal. We do not assume that bicycle users cause crime, that offenders travel by bicycle, or that bicycle flow measures the whole ambient population. The empirical question is whether lagged flow contains incremental forecast-time information about subsequent property-crime counts and whether an estimated gain can be distinguished from finite-sample artifacts.

1.2. From Integer-Valued Uncertainty to Calibrated Evidence

Conditional entropy provides a natural measure of the uncertainty remaining after the forecast-time information set has been observed [18,19]. Existing entropy-based prediction bounds, however, are mainly formulated for continuous outcomes or classification losses. A continuous Gaussian entropy–variance argument cannot be transferred to integer-valued crime counts by simply replacing differential entropy with Shannon entropy. The prediction error lies on an integer lattice, and the relevant maximum-entropy envelope under a second-moment constraint is discrete.

To address this support mismatch, we define the Discrete Entropy Lower Bound (DELB). DELB maps conditional Shannon entropy through the inverse integer-lattice maximum-entropy envelope to a model-independent lower bound on the expected squared one-step forecast error, $\mathbb{E}[(X_t - \hat{X}_t)^2]$, for forecast-time causal, or non-anticipating, integer-valued predictors. Here, “causal” refers only to the temporal admissibility of information available when a forecast is issued; it does not denote identification of a causal effect. The envelope inversion is exact for the stated integer-lattice maximum-entropy problem, although equality with the optimal integer-predictor risk requires the stated attainability conditions. The framework also includes a predictor-dependent refinement and separates

two sources of nonattainment: residual dependence of the prediction error on the available history and mismatch between the error distribution and the moment-matched discrete Gaussian.

The discrete-Gaussian maximum-entropy law and lattice entropy corrections are established results. Here, they are adapted to non-anticipating integer prediction and linked to predictor-specific attainability conditions, incremental information analysis, and finite-sample calibration.

A population ordering addresses only the first part of the applied problem. Adding a discretized mobility variable refines the baseline state space, often creating sparse or singleton cells. Conditional-entropy and conditional mutual information (CMI) estimators can then have different finite-sample errors in the baseline and augmented state tables [30,31]. A positive estimated CMI, or a lower estimated DELB, may consequently arise even when the added variable has no population information value. A randomization reference is also conditioned on its transformation design and need not be centered at zero when that design changes support occupancy. Permutation inference requires an explicit transformation design [35]. CMI-based conditional-independence methods use, respectively, nearest-neighbor estimation [33], mixed-data estimation [32], and nearest-neighbor sampling [34].

The evaluation must therefore distinguish four objects: the population DELB and CMI estimands, their finite-sample estimates, evidence relative to a prespecified randomization reference, and realized out-of-sample prediction error. We connect these layers using known-truth simulations, support diagnostics, block resampling, temporal and spatial randomization, circular-shift references, and matched prediction models. This design does not turn a transformation-specific diagnostic into a causal or exact conditional-independence test. Its purpose is to determine whether the observed statistic is distinguishable from finite-sample behavior under each explicit reference design.

1.3. Research Questions and Contributions

The study addresses three linked research questions:

1. *How can conditional uncertainty be translated into a discrete-form, model-independent MSE lower bound for non-anticipating integer-valued prediction?*
2. *How can a positive CMI estimate be separated from sparse-state estimation bias and a transformation-specific randomization reference?*
3. *Under this calibrated framework, what evidence does lagged bicycle mobility provide about future urban crime counts, and when is that information realized by fitted prediction models?*

The study makes three contributions. First, it organizes established integer-lattice maximum-entropy results into an explicit DELB framework for non-anticipating integer prediction, together with attainability and nonattainment diagnostics. Second, it develops a finite-sample procedure for assessing whether an apparent reduction in the bound can be distinguished from sparse-state estimation effects. Third, it applies the procedure to harmonized crime and shared-bicycle data from three cities and compares the estimated bounds with matched held-out prediction errors.

Adding lagged bicycle flow lowers the estimated DELB in both city-level analysis populations. In the primary bicycle-covered, grid-identity specification, however, the observed CMI is not distinguishable from the prespecified temporal, spatial, circular-shift, or baseline-state-matched references after calibration. In a lower-cardinality sensitivity, only New York City shows a small matched-reference difference after multiplicity adjustment; this finding qualifies the primary non-rejection and demonstrates the dependence of finite-sample resolution on state design. Realized prediction changes are model- and city-dependent. The empirical application therefore serves as a calibration case study

in which the estimated DELB reduction, randomization-based evidence, and predictive performance are evaluated separately.

Section 2 reviews urban crime forecasting, mobility data in crime analytics, and information-theoretic prediction limits. Section 3 presents the DELB methodology. Section 4 defines the multicity data and experimental design. Section 5 first summarizes the multicity application and then reports estimated DELB changes, finite-sample behavior, randomization calibration, and realized prediction. Section 6 discusses practical implications, inferential scope, and limitations.

2. Related Work

2.1. Urban Crime Forecasting

Crime forecasting seeks to anticipate future incident counts, hotspot status, or event risk at operationally relevant locations and times. This variation in target is accompanied by variation in scale and model design. Reviews of spatial forecasting [1] and multiscale methods [2] frame the geographic and temporal choices, while a machine- and deep-learning review [3] surveys the algorithmic choices. Their different scopes help explain why reported forecast scores are difficult to compare without a common target and evaluation design.

The models themselves show how the field has addressed those choices. Urban metrics enter statistical learning as contextual predictors [4]. Attentive recurrent networks model temporal structure [6]; a spatiotemporal deep-learning model [5] and geographically informed risk models [7] extend the representation of spatiotemporal dependence. Multiscale hotspot mapping [8] and graph-based learning of correlations among crime types [9] address different forms of spatial and cross-category structure. More recent transformer prediction [10] and transfer with multi-source fusion [11] broaden the model and data choices; a survey of crime event analysis documents the range of fusion strategies [12]. Across these families, performance is reported through hotspot accuracy, prediction-accuracy indices, precision and recall, F1 scores, and count losses such as mean absolute error (MAE), root mean squared error (RMSE), and mean squared error (MSE). Reported performance depends not only on the forecasting algorithm but also on the target definition, spatial and temporal resolution, validation period, comparison model, and distribution shift. Multidensity and multiscale studies further show that changing the unit of analysis can alter the forecasting problem itself [8,13], while recent evaluation frameworks identify limited standardization and cross-design comparability as persistent concerns [14]. Closely related urban-computing studies illustrate the same design dependence: recent work models long-horizon traffic with contextual knowledge [15], adapts shared-bicycle demand models to spatial and temporal heterogeneity [16], and provides a unified toolbox for constructing and comparing spatiotemporal prediction services [17]. Thus, a lower test error establishes that one specification performs better under a particular evaluation design; it does not establish how close that specification is to an information-constrained prediction limit. The unresolved issue is not a shortage of forecasting models, but the absence of a model-independent way to assess whether an added information source changes the attainable error benchmark for an integer-valued target.

2.2. Mobility Data in Crime Analytics

Routine activity theory and ambient-population research imply that crime opportunities depend on the changing presence of potential targets, offenders, and guardians, not only on residential population [36,37]. Song et al. relate daily mobility flows to crime targets [38]; Kadar and Pletikosa use large-scale mobility data for long-term prediction [39]; and Kadar et al. use location-platform flows to examine crime pattern theory [40].

Wu et al. combine mobility flows with deep learning for short-term prediction [41]; Rumens et al. test mobile-phone denominators and forecasts [42]; and Schertz et al. examine neighborhood street activity [43]. A recent systematic review documents both the rapid growth and the heterogeneous uses of mobile-phone data in crime research [44]. At finer scales, mobile-phone occupancy has been related to outdoor robbery and theft [45], and a four-city forecasting study reports gains from combining mobility, crime history, and sociodemographic features [46].

This evidence supports testing mobility as added information, but mobility sources are not interchangeable. Digital traces differ in population coverage, spatial support, temporal rhythm, directionality, and mode of travel [47,48]. Mobile-location data can also exhibit scale- and time-dependent representation bias [49]. More broadly, a recent survey of foundation models for the Internet of Things identifies context awareness, efficiency, and safety as distinct criteria for incorporating sensor information [50]. We therefore treat shared-bicycle records as a high-frequency, spatially resolved, but mode-specific proxy for observed public-space activity [51,52], not as a census of ambient population. Predictive information also does not establish a causal effect: recent instrumental-variable evidence illustrates that predictive association and causal identification can lead to different conclusions [53].

Most mobility–crime studies evaluate association patterns or changes in fitted prediction scores. These questions remain useful, but they do not separate the information value of a mobility source under a specified target and information set from the ability of a particular model to exploit it. Nor do they generally diagnose whether an apparent gain is induced by sparse states after mobility variables are discretized and joined to a fine-scale crime panel. This gap motivates the matched baseline–augmented information sets and finite-sample calibration used in the present study.

2.3. Information-Theoretic Prediction Limits

Conditional entropy measures uncertainty after admissible information has been observed [18,19]. Entropy and Fano-type arguments have been used to study potential predictability in human mobility [20]; conditional differential entropy has also been used to bound traffic-forecasting performance [21]. Fang et al. derive general entropy-based variance bounds for continuous estimation and prediction errors [22]. Recent work has extended this continuous line to time-series complexity ranking and entropy-based lower bounds on predictive representation dimension [23,24]. These studies separate achieved performance from an information-constrained limit, but their target spaces and bound objects do not directly provide the integer-lattice MSE lower bound required here.

Crime counts and many other operational targets are integer valued. A continuous Gaussian argument cannot be made discrete by replacing differential entropy with Shannon entropy: the error lies on a lattice, and its maximum-entropy law under a second-moment constraint is a discrete Gaussian. Integer maximum-entropy and lattice-correction results are established ingredients [25,26]; general Bayes-risk, rate–distortion, and Ziv–Zakai theories also show that lower bounds on estimation risk are not new as a class. Shannon formulated rate–distortion bounds [27]; Chen et al. studied Bayes-risk bounds [28]; and Jeong et al. examined Ziv–Zakai bounds on MMSE [29]. What remains underdeveloped for the present problem is an operational framework for non-anticipating integer prediction that connects an explicit lattice MSE floor to the predictor used, states necessary and sufficient attainability conditions, diagnoses why the floor is not attained, values added information, and calibrates its finite-sample estimate.

We refer to the MSE lower bound obtained by inverting the exact integer-lattice maximum-entropy envelope as the *Discrete Entropy Lower Bound* (DELB). The theoretical

package combines a universal floor, a predictor-dependent refinement, strict attainability conditions, and a split of nonattainment into residual history dependence and error-shape mismatch. To our knowledge, these components have not previously been connected to incremental information valuation and finite-sample calibration in a single operational workflow for integer-valued urban forecasting. The contribution lies in this integration and specialization, not in the discrete-Gaussian extremizer, the lattice correction, or a new primitive divergence identity.

Table 1 makes the distinction operational by comparing target, loss, support, and the role of each framework in the present study.

Table 1. Operational comparison of prediction-risk lower-bound frameworks. The DELB contribution is an integrated specialization and calibration workflow, not a new discrete-Gaussian maximum-entropy result.

Framework	Target	Loss or quantity	Support	Relationship to this study
Fano inequality [19,63]	Finite classes	Hamming error	Finite label set	Secondary classification-loss comparison; not an MSE bound
Fang et al. [22]	Continuous estimation or prediction	Variance/MSE	Continuous	Continuous analogue; it does not directly handle an integer-lattice error
Massey/Rioul [25,26]	Integer-valued variable	Variance or second moment	Integer lattice	Supplies the established maximum-entropy and lattice-correction basis
Bayes-risk/rate-distortion bounds [27,28]	General decision problem	General loss or distortion	General	Broader than DELB, but often less explicit for the present count setting
Present DELB framework	Non-anticipating integer prediction	MSE	Integer lattice	Integrates an explicit floor, predictor refinement, attainability and nonattainment diagnostics, added-information ordering, and finite-sample calibration

Across these three literature streams, a specific applied methodological gap remains. Crime forecasting provides increasingly flexible models, and mobility studies provide increasingly detailed urban signals, whereas existing entropy bounds do not supply a finite-sample-calibrated, integer-lattice benchmark for determining whether such a signal changes the information-constrained lower bound under a specified count target and information set. DELB and the accompanying calibration workflow address this intersection. The comparison is intentionally between information sets rather than between algorithm families, and between predictive evidence rather than causal effects.

3. Methodology: Discrete Entropy Lower Bound

3.1. General Integer-Valued Prediction Setting

Let $\{X_t\}_{t \in \mathbb{Z}}$ be an integer-valued stochastic process,

$$X_t \in \mathbb{Z}.$$

(1)

For an m -step forecast, $m \geq 1$, let $\mathcal{F}_{t-m}$ denote the sigma-algebra containing exactly the observations available when the forecast is issued. A non-anticipating integer-valued predictor is selected from the explicitly restricted decision class

$$\mathcal{A}_{\mathbb{Z}}(\mathcal{F}_{t-m}) = \left\{ f(\mathcal{F}_{t-m}) : f \text{ is measurable, } f(\mathcal{F}_{t-m}) \in \mathbb{Z}, \mathbb{E}[(X_t - f(\mathcal{F}_{t-m}))^2] < \infty \right\},$$

(2)

and $\hat{X}_t \in \mathcal{A}_{\mathbb{Z}}(\mathcal{F}_{t-m})$. Its prediction error and mean squared error (MSE) are

$$E_t = X_t - \hat{X}_t, \quad D_t^{(m)} = \mathbb{E}[E_t^2]. \quad (3)$$

Here, E_t is the integer-valued forecast error, X_t is the realized count, $\hat{X}_t$ is the forecast issued from information available at time $t - m$, and $D_t^{(m)}$ is the expected squared forecast error at horizon m . Thus, every MSE bound below applies to the prediction error $X_t - \hat{X}_t$, not to the count itself or to a parameter estimator. All logarithms have base 2; entropy is measured in bits.

Assumption 1. For every time and horizon considered, $H(X_t \mid \mathcal{F}_{t-m}) < \infty$ and $\mathbb{E}[E_t^2] < \infty$. Forecast information sets are temporally admissible and nested: $\mathcal{F}_{t-m} \subseteq \mathcal{F}_{t-\ell}$ whenever $1 \leq \ell < m$.

For a discrete random variable V with mass function $p(v)$,

$$H(V) = - \sum_v p(v) \log_2 p(v), \quad (4)$$

Here, $p(v) = \Pr(V = v)$ is the probability mass at outcome v . and conditional mutual information (CMI) satisfies

$$I(V; W \mid Y) = H(V \mid Y) - H(V \mid Y, W). \quad (5)$$

Here, V is the target, W is the added information, and Y is the information already available for conditioning. The distinction between a population quantity such as $H(X_t \mid \mathcal{F}_{t-m})$ and its finite-sample estimate is essential: the former defines the lower bound below, whereas the latter is an inferential object studied in Sections 4.6 and 5.

3.2. Discrete Entropy Prediction-Error Lower Bound and Predictor-Dependent Refinement

Continuous entropy–variance arguments rely on the fact that a Gaussian maximizes differential entropy [22]. An integer-valued error lies on a lattice, so differential entropy cannot be replaced mechanically by Shannon entropy. The relevant extremal object is the integer-lattice maximum-entropy envelope [25].

Definition 1 (Integer maximum-entropy envelope). For $D \geq 0$, define

$$\mathcal{H}_{\mathbb{Z}}(D) = \sup_{\substack{P_Z: Z \in \mathbb{Z} \\ \mathbb{E}[Z^2] \leq D}} H(Z), \quad (6)$$

Here, P_Z ranges over distributions of integer-valued random variables Z , and D is an upper bound on their second moment. with generalized inverse

$$\mathcal{H}_{\mathbb{Z}}^{-1}(h) = \inf\{D \geq 0 : \mathcal{H}_{\mathbb{Z}}(D) \geq h\}. \quad (7)$$

For $D > 0$, the envelope is attained by the centered discrete Gaussian

$$q_{\lambda}(k) = \frac{\exp(-\lambda k^2)}{\Theta(\lambda)}, \quad \Theta(\lambda) = \sum_{j \in \mathbb{Z}} \exp(-\lambda j^2), \quad k \in \mathbb{Z}, \quad (8)$$

Here, k is an integer error value, $\lambda > 0$ controls dispersion, and $\Theta(\lambda)$ normalizes the mass function. where $\lambda > 0$ is selected so that

$$D = D(\lambda) = -\frac{d}{d\lambda} \ln \Theta(\lambda). \quad (9)$$

Its entropy is

$$H_\lambda = \frac{\lambda D(\lambda) + \ln \Theta(\lambda)}{\ln 2}. \quad (10)$$

Equations (9) and (10) define the one-dimensional numerical inversion used in this study.

Theorem 1 (Discrete entropy prediction-error lower bound and predictor-dependent refinement). Under Assumption 1, every non-anticipating integer-valued predictor satisfies

$$D_t^{(m)} \geq \mathcal{H}_{\mathbb{Z}}^{-1}[H(X_t | \mathcal{F}_{t-m}) + I(E_t; \mathcal{F}_{t-m})] \quad (11)$$

$$\geq \mathcal{H}_{\mathbb{Z}}^{-1}[H(X_t | \mathcal{F}_{t-m})]. \quad (12)$$

The second line is the discrete entropy prediction-error lower bound (DELB).

Equation (12) gives the population-level answer to RQ1: target uncertainty and the integer lattice impose an MSE lower bound independently of the forecasting algorithm. Equation (11) is stronger for a particular predictor because it retains the residual information quantified by $I(E_t; \mathcal{F}_{t-m})$. The proof and its boundary cases accompany the article together with the closed-form, temporal, stationary, and vector consequences.

To distinguish the DELB from minimum decision risk, define the optimal risk within this same non-anticipating integer-valued predictor class by

$$R_{\mathbb{Z}}^*(\mathcal{F}) = \inf_{\hat{X} \in \mathcal{A}_{\mathbb{Z}}(\mathcal{F})} \mathbb{E}[(X - \hat{X})^2]. \quad (13)$$

For every admissible integer-valued predictor,

$$L_{\text{DELB}}(\mathcal{F}) \leq R_{\mathbb{Z}}^*(\mathcal{F}) \leq \mathbb{E}[(X - \hat{X})^2]. \quad (14)$$

Thus, the exactness of the DELB concerns the discrete entropy envelope and its inversion. It does not make the DELB identical to optimal risk. If the infimum in Equation (13) is attained, equality $L_{\text{DELB}} = R_{\mathbb{Z}}^*$ holds if and only if a minimizing predictor satisfies the attainability conditions below. Without attainment of the infimum, the hierarchy remains valid, but equality of numerical infima must not be interpreted as the existence of a predictor that attains the DELB.

3.3. Attainability and the Two-Slack Decomposition

Proposition 1 (Exact entropy-domain attainability decomposition). Abbreviate $\mathcal{F} = \mathcal{F}_{t-m}$, $E = E_t$, and $D = \mathbb{E}[E^2]$. For $D > 0$, let q_D be the centered discrete Gaussian with second moment D ; for $D = 0$, let q_0 be the point mass at zero. Then

$$\mathcal{H}_{\mathbb{Z}}(D) - [H(X_t | \mathcal{F}) + I(E; \mathcal{F})] = D_{\text{KL},2}(P_E \| q_D), \quad (15)$$

$$\mathcal{H}_{\mathbb{Z}}(D) - H(X_t | \mathcal{F}) = \underbrace{I(E; \mathcal{F})}_{\text{history-dependence slack}} + \underbrace{D_{\text{KL},2}(P_E \| q_D)}_{\text{shape-mismatch slack}}, \quad (16)$$

where $D_{\text{KL},2}$ is the Kullback–Leibler (KL) divergence in bits.

The predictor-specific bound is attained if and only if $P_E = q_D$. The universal DELB is attained if and only if $P_E = q_D$ and

$$I(E_t; \mathcal{F}_{t-m}) = 0. \quad (17)$$

The error must be moment-matched discrete Gaussian and independent of the admissible history. Zero correlation or second-order residual whiteness is insufficient. The two terms in Equation (16) diagnose different population sources of nonattainment; neither is the empirical Predictability Gap introduced in Section 4.14.

Detailed comparisons with neighboring information-theoretic bounds and the secondary corollaries are kept outside the main text so that it can focus on the implementable bound, its attainability diagnostics, and added-information ordering.

Remark 1 (Boundary and loss conditions). *Because $\mathcal{H}_{\mathbb{Z}}^{-1}(0) = 0$, information that determines X_t almost surely produces a zero DELB. Conditionally irrelevant side information leaves it unchanged. Shannon entropy alone does not define an MSE lower bound without integer support and squared-error loss; the lattice envelope is an essential component of Theorem 1.*

3.4. Added Side Information and CMI Ordering

Let $\mathcal{F}_t^0$ be a baseline forecast-time information set and let S_t be additional side information observed when the forecast is issued. Define

$$\mathcal{F}_t^S = \mathcal{F}_t^0 \vee \sigma(S_t). \quad (18)$$

Here, $\mathcal{F}_t^0$ is the baseline forecast-time information, S_t is the added observation, $\sigma(S_t)$ is the sigma-algebra it generates, and $\vee$ denotes the information generated jointly by its arguments. The population uncertainty reduction is exactly

$$\Delta H_t^S = H(X_t | \mathcal{F}_t^0) - H(X_t | \mathcal{F}_t^S) = I(X_t; S_t | \mathcal{F}_t^0) \geq 0. \quad (19)$$

Because the inverse envelope is nondecreasing, the corresponding universal floors

$$L_t^0 = \mathcal{H}_{\mathbb{Z}}^{-1}[H(X_t | \mathcal{F}_t^0)], \quad L_t^S = \mathcal{H}_{\mathbb{Z}}^{-1}[H(X_t | \mathcal{F}_t^S)] \quad (20)$$

satisfy

$$L_t^S \leq L_t^0. \quad (21)$$

Equations (19)–(21) define the population estimand for RQ2 and RQ3. They do not imply that an estimated CMI is unbiased, that a design-conditioned randomization distribution is centered at zero, or that a finite-sample predictor will realize the population reduction. These are separate empirical questions. Deterministic coarsening $Q(S_t)$ cannot increase conditional information:

$$I(X_t; Q(S_t) | \mathcal{F}_t^0) \leq I(X_t; S_t | \mathcal{F}_t^0). \quad (22)$$

The Supplementary Materials give the complete proofs, place the DELB relative to established information-theoretic bounds, and state the supplementary classification-loss result. Section 4.6 maps the generic objects $(X_t, \mathcal{F}_t^0, S_t)$ to crime counts, admissible controls, and lagged bicycle mobility.

The four main-text components have distinct operational roles: the universal DELB defines the benchmark, the predictor-dependent term and two-slack decomposition diagnose nonattainment, and CMI orders information sets at the population level. Section 4.6 sup-

plies the finite-sample estimation, calibration, and matched-prediction workflow needed to apply these quantities.

4. Data and Experimental Design

4.1. Study Design and Evidence Map

The empirical analysis covers Washington, DC, New York City, and Vancouver from 1 January 2020 through 31 December 2022. These cities were selected because each provides geocoded incident records and a large public bicycle-system archive over a common period. Crime records came from the Washington, DC Metropolitan Police Department [54], the New York City Police Department [55], and the Vancouver Police Department [56]. Bicycle trips came from Capital Bikeshare in Washington, DC [57], Citi Bike in New York City [58], and Mobi by Rogers in Vancouver [59].

Figure 1 links the three research questions to the corresponding evidence. RQ1 concerns a population limit and is answered analytically. RQ2 asks whether a finite-sample entropy difference can be separated from sparse-state bias and a design-conditioned null. RQ3 applies the resulting calibration to lagged bicycle mobility and crime counts, followed by held-out evaluation of the predictive consequence.

Figure 2 summarizes data processing. Crime and bicycle records were first standardized as event-level data. They were then aggregated to complete daily panels on a common 1 km grid. The upper branch estimates conditional entropy, conditional mutual information (CMI), and the DELB; CMI is evaluated against design-matched null references. The other branch evaluates matched integer-valued predictors on held-out observations. The Predictability Gap combines the estimated DELB with their held-out mean squared error (MSE), after information sets and evaluation periods have been aligned.

4.2. Crime Events and Cross-City Harmonization

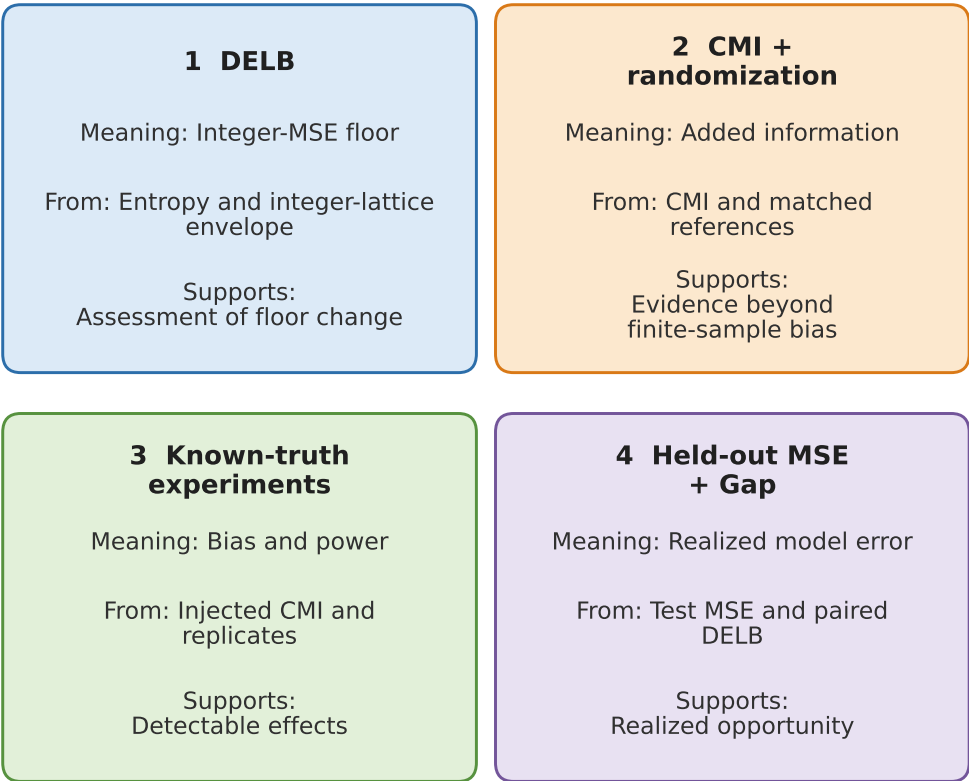
The raw crime files were harmonized at the event level. We retained the authoritative standardized field `crime_type_unified` and did not reclassify it in later experiments. The analytic outcome combines three mutually exclusive categories: property theft, vehicle theft, and burglary. In this harmonization, *vehicle theft* means theft of a general motor vehicle, not theft of a bicycle. *Property theft* covers non-burglary taking, including other theft and theft from vehicles (and, in Vancouver, source-labeled bicycle theft), whereas *burglary* requires unlawful entry into a building or other fixed premise. These categories are pooled because they are the mutually exclusive property-crime outcomes that can be mapped consistently across all three cities. Bicycle flow is not treated as a mode used in the offense; it is a lagged indicator of local street activity that may be informative for both mobile theft opportunities and activity near fixed premises. Events were assigned to their source file's calendar year; 35 Washington, DC records whose event dates fell outside that year were placed in a reason-coded exclusion table. Stable event identifiers, source file hashes, source row numbers, original categories, timestamps, and coordinates were retained for auditability.

Standardization produced 503,157 valid crime events. After applying the prespecified spatial domain described below, 503,128 events entered the complete analytic panels. The primary prediction target, $C_{c,i,t}$, is the unbinned daily count of all three harmonized categories in city c , grid cell i , and local calendar day t . Category-specific counts are used only in the heterogeneity analysis.

4.3. Bicycle Trips and Endpoint Flows

Schemas differed across the three systems and changed during the study period. We standardized them to a common trip interface with start and end times, station identifiers,

Evidence layers in the analysis

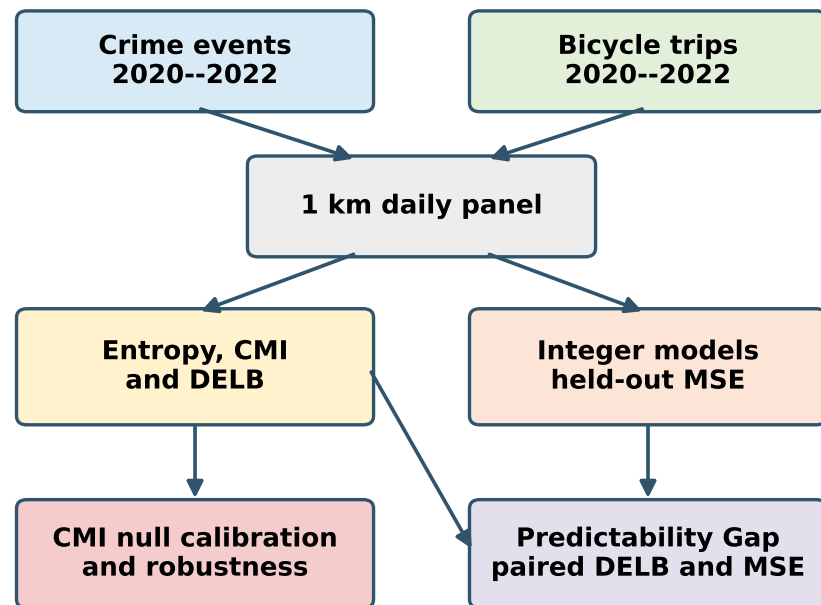


The four layers answer related but different questions.

Figure 1. Evidence guide for the analysis. Each panel identifies an evidence object, its practical meaning, how it is calculated, and the conclusion it can support. The numbered panels form a sequence rather than interchangeable significance tests. CMI denotes conditional mutual information, and DELB denotes the Discrete Entropy Lower Bound.

station names, endpoint coordinates, coordinate provenance, source identifiers, and quality-control flags. Input files were processed in chunks, and month assignment was determined by trip start time. Blank rows and trips assigned to a different source month were retained in reason-coded exclusion tables. Official ride identifiers were required to be unique after deterministic first-record retention. The Vancouver files contained 217 repeated complete row signatures among 2,256,222 source rows (0.00962%) but supplied no unique ride identifier. Because distinct trips can share every recorded attribute, these rows were retained with an explicit flag; even treating all 217 as duplicates would affect at most 434 endpoint contributions. We also ran the requested source-row-specific sensitivity: retaining the first occurrence of each complete raw-row signature and removing only the 217 subsequent rows removed 353 eligible endpoint contributions from the fixed Vancouver analysis domain. The primary Miller–Madow CMI changed from 0.142329 to 0.142416 bits, and the exact DELB reduction changed from 0.091906 to 0.091956 MSE units. The original training-period cut points, covered domain, crime observations, and temporal split were unchanged. This numerically small change does not establish that the 217 records were duplicate journeys; the source provides no ride identifier to decide that. As a deliberately

Information-theoretic and predictive analysis workflow



All bicycle variables are lagged and restricted to information available at forecast time.

Figure 2. Data-processing and matched-analysis workflow. CMI enters the null-calibration analysis; the Predictability Gap combines DELB and held-out model MSE. All bicycle variables are lagged and restricted to information observed when the one-day-ahead forecast is issued.

over-inclusive sensitivity check, we removed all 3,880 repeated standardized trip signatures (6,806 in-domain endpoint contributions), although some may be distinct journeys. Re-estimating the Vancouver primary panel changed the Miller–Madow CMI from 0.142329 to 0.142428 bits and the exact DELB reduction from 0.091906 to 0.091964 MSE units. This stress test supports numerical stability of the reported point estimate; it does not establish which of the 217 repeated complete source rows represent genuinely duplicated rides.

Source coordinates were used when valid. Missing coordinates in Washington, DC and New York were recovered from robust station-month medians and temporally adjacent observations. Vancouver endpoints were matched first to the official station identifier and then to a unique normalized station name in the supplied official station file. Workshop, service, testing, yard, storage, production, temporary, and valet nodes were marked non-public. Unresolved endpoints were preserved in the standardized trip table but were ineligible for spatial flow at that endpoint; a trip could still contribute its other resolved public endpoint.

Coordinate provenance was audited over all retained trip endpoints. In Washington, DC, 93.532895% were recorded directly in the trip source, 6.361190% were resolved deterministically from a station reference, and 0.105915% remained unresolved. The corresponding New York shares were 99.906369%, 0.000006%, and 0.093594%; a further 0.000031% were identified as non-public nodes. Vancouver trip files did not contain endpoint coordinates: 95.513437% of endpoints were matched deterministically to the official station snapshot, 4.322067% remained unresolved, and 0.164496% were non-public nodes. No crime-event coordinates were statistically imputed. These operations are station-key lookups, not model-based imputation: they do not use crime outcomes, calendar outcomes, or fitted values. Unresolved and non-public endpoints are excluded from flow at that endpoint,

and the other endpoint remains eligible. The crime-category crosswalk and the full mutually exclusive coordinate audit are consolidated in the Supplementary Materials under “Empirical Harmonization and Audit Details.” We also performed a same-city coordinate-source sensitivity for Washington, DC and New York City. Holding the crime observations, grid, calendar splits, and training-period bicycle-covered domain fixed, we removed only flow contributions from bicycle endpoints located through station-reference matching, re-estimated the primary CMI and DELB, and repeated the three randomization comparisons. This is a post hoc data-provenance sensitivity, not a new prespecified primary analysis. Vancouver cannot enter a comparable source-only contrast because its trip files supply no endpoint coordinates: removing the official station snapshot would remove almost all bicycle flow by file-schema design. This dependence on deterministic station metadata remains a data-quality limitation.

For each grid-day, departures and arrivals define $B_{c,i,t}^{\text{out}}$ and $B_{c,i,t}^{\text{in}}$, with total endpoint flow

$$B_{c,i,t}^{\text{total}} = B_{c,i,t}^{\text{out}} + B_{c,i,t}^{\text{in}}.$$

Here, $B_{c,i,t}^{\text{out}}$ and $B_{c,i,t}^{\text{in}}$ are the numbers of resolved public departures and arrivals, respectively, for city c , grid i , and day t , and $B_{c,i,t}^{\text{total}}$ is their sum. The primary mobility variable is the one-day lag of total flow. This measure captures observable activity at both ends of a trip without including the deterministically redundant total and net flow together with inflow and outflow in the same entropy state. Standardization retained 87,208,600 trips; 86,269,976 trips had at least one flow-eligible endpoint inside the analytic grids.

4.4. Spatial and Temporal Panel Construction

Coordinates were projected in the appropriate city coordinate reference system and assigned to deterministic 1 km square cells anchored to the projected coordinate-system origin. Two training-period-frozen analysis populations were prespecified for city-level point estimation. The *deployment population* uses the complete crime-supported domain, containing every cell with at least one crime event during 2020–2021. The *bicycle-covered population* uses the bicycle-covered domain, the subset of those cells with positive total bicycle flow during the same training period. The former asks about the added source for the complete target task; the latter asks about its value where the mobility channel is observed. Neither definition uses a 2022 record. Bicycle endpoints outside the complete crime-supported domain remained in reconciliation outputs but were not used in either analytic population.

Every retained grid was crossed with all 1096 calendar days, including zero-crime and zero-flow days. The resulting complete panel contains 1,241,768 grid-day observations across 1133 cells. The bicycle-covered panel is a fixed spatial subset of this complete panel; it was not reconstructed from the set of nonzero-flow days. Lag construction removed the first day of each grid only when a lagged analysis was performed.

The temporal split was fixed before model estimation: 2020–2021 for training, January–June 2022 for validation, and July–December 2022 for testing. Spatial eligibility, bicycle coverage, and bicycle discretization thresholds were determined from training records only. Model hyperparameters were selected using training and validation data only. The test period defined neither the domains nor any threshold, model setting, or eligibility decision. All predictors used for a forecast were available at issuance: no concurrent or future crime or mobility record entered the information sets. Table 2 makes this timing explicit, and Table 3 summarizes the complete panels.

Table 2. Variables, analysis units, and forecast-time availability. “Yes, lagged” indicates that only information from the previous aggregation period was used for one-period-ahead prediction.

Variable	Meaning and role	Spatial unit	Temporal unit	Available at forecast time
Crime count	Daily incidents in the three harmonized crime categories; un-binned prediction target	1 km grid	Day	Target, no
Lagged crime	Previous-period crime count, discretized into four levels (0, 1, 2, and 3+); baseline state	1 km grid	Previous day	Yes, lagged
Bicycle inflow	Arrivals at resolved public endpoints; mobility primitive	1 km grid	Previous day	Yes, lagged
Bicycle outflow	Departures from resolved public endpoints; mobility primitive	1 km grid	Previous day	Yes, lagged
Total bicycle flow	Sum of inflow and outflow, discretized using training-period thresholds; added-data state	1 km grid	Previous day	Yes, lagged
Calendar state	Day of week, meteorological season, and public-holiday status; baseline state	City/grid	Forecast day	Yes
Grid identifier	Fixed spatial state defined before validation and testing; baseline state	1 km grid	Fixed	Yes

Table 3. Descriptive statistics for the complete 1 km grid–day panels, 2020–2022.

City	Grids	Grid–days	Crime events	Bike trips	Mean crime	Mean bike flow	Zero crime
Washington, DC	186	203,856	71,397	7,620,562	0.350	72.95	77.0%
New York City	806	883,376	359,284	76,418,121	0.407	172.82	79.7%
Vancouver	141	154,536	72,447	2,231,293	0.469	27.37	75.3%

Crime was sparse at the grid–day scale: zero counts represented 75.3–79.7% of observations. New York contributed the largest system and the highest mean daily bicycle endpoint flow, whereas Vancouver had the highest mean property-crime count per grid–day. We report city-specific quantities and keep the city samples separate.

4.5. Descriptive Analysis

Monthly and spatial summaries were computed before entropy estimation. At the monthly level, Spearman correlations between crime and bicycle flow were 0.505 in Washington, DC, 0.603 in New York, and 0.221 in Vancouver. Across grids, the corresponding correlations of three-year totals were 0.637, 0.651, and 0.633. The top decile of grids accounted for 43.9, 54.3, and 54.1% of crime but 67.5, 89.0, and 79.7% of bicycle flow, respectively. These are descriptive associations and do not condition on crime history or calendar state.

The main text proceeds directly to the DELB estimand and its finite-sample validation; the descriptive spatial-concentration and monthly-series figures are grouped with the secondary empirical analyses identified at the end of the Methods.

4.6. Evaluation Design Overview

Analysis proceeded in the order of the research questions. We first operationalized the target and information sets, then evaluated the numerical DELB and examined finite-sample CMI behavior with known-truth simulation and support diagnostics, estimated city and local point values, calibrated them against three randomization references, and only then compared matched held-out predictions. Fano, crime-type, and alternative-specification analyses served as secondary checks.

4.7. Targets and Operational Information Sets

For the application, the generic target in Section 3 is

$$X_t \mapsto C_{c,i,t} \in \mathbb{Z}_{\geq 0}, \quad (23)$$

Here, $C_{c,i,t}$ is the unbinned property-crime count for city c , grid cell i , and local day t .

Bicycle departures and arrivals are

$$B_{c,i,t}^{\text{out}} = \text{departures from } (c, i, t), \quad B_{c,i,t}^{\text{in}} = \text{arrivals at } (c, i, t), \quad (24)$$

with primitive vector

$$\mathbf{B}_{c,i,t} = (B_{c,i,t}^{\text{out}}, B_{c,i,t}^{\text{in}})^\top, \quad (25)$$

total and net flow

$$B_{c,i,t}^{\text{total}} = B_{c,i,t}^{\text{out}} + B_{c,i,t}^{\text{in}}, \quad (26)$$

$$B_{c,i,t}^{\text{net}} = B_{c,i,t}^{\text{in}} - B_{c,i,t}^{\text{out}}, \quad (27)$$

and lag block

$$\mathbf{B}_{c,i,t}^{(q)} = (\mathbf{B}_{c,i,t-1}, \dots, \mathbf{B}_{c,i,t-q}). \quad (28)$$

Total and net flow are deterministic functions of the primitive vector and are not entered redundantly in one entropy state.

For one-day-ahead prediction, the target remained the raw nonnegative integer count. Only conditioning variables were discretized. The previous day's crime count was mapped to $\{0, 1, 2, \geq 3\}$. Within each city, the positive values of lagged total bicycle flow were divided at training-period terciles; zero flow formed a separate state. This produced four bicycle states: $\{\text{zero, low, medium, high}\}$, and the fitted cut points were reused unchanged in validation and testing.

The baseline information state was

$$\mathcal{S}_{c,i,t}^0 = (i, Q_C(C_{c,i,t-1}), d_t, s_t, h_t), \quad (29)$$

where d_t , s_t , and h_t denote day of week, meteorological season, and public-holiday status. The bicycle-augmented state was

$$\mathcal{S}_{c,i,t}^B = (\mathcal{S}_{c,i,t}^0, Q_B(B_{c,i,t-1}^{\text{total}})). \quad (30)$$

Only one component differs between the matched states: lagged bicycle flow. The design excludes concurrent or future mobility. In sigma-algebra notation,

$$\mathcal{F}_{c,i,t}^0 = \sigma(\mathcal{S}_{c,i,t}^0), \quad (31)$$

$$\mathcal{F}_{c,i,t}^B = \mathcal{F}_{c,i,t}^0 \vee \sigma\{Q_B(\mathbf{B}_{c,i,t}^{(q)})\}. \quad (32)$$

Weather and land-use covariates were not used in the reported primary analyses and are not claimed as controls.

Three additional sensitivities were frozen before comparison with their results. First, a lower-cardinality specification replaced grid identity with within-city quintiles of the training-period mean crime count; all remaining baseline variables were unchanged. Second, calendar month replaced four-level meteorological season. Third, the bicycle state used zero plus positive-flow quintiles. A day-of-year control was not added because its 365 levels would further multiply the documented sparse conditioning space; calendar

month was used as the lower-cardinality sensitivity for finer seasonal adjustment. Each specification recomputed support metrics, Miller–Madow CMI, DELB change, and the seven-day baseline-state-matched reference. Routine sensitivities used 199 surrogates; the key New York lower-cardinality result was confirmed with 999. Plus-one upper-tail p -values for the three-city lower-cardinality family were adjusted by Benjamini–Hochberg.

4.8. Crime-Count and Bicycle-Information Specialization

Using the variables defined above, the baseline entropy and bounds are

$$H_{c,i,t}^0 = H(C_{c,i,t} \mid \mathcal{F}_{c,i,t}^0), \quad (33)$$

$$L_{c,i,t}^0 = \mathcal{H}_{\mathbb{Z}}^{-1}[H_{c,i,t}^0], \quad (34)$$

$$\mathbb{E}[(C_{c,i,t} - \hat{C}_{c,i,t})^2] \geq L_{c,i,t}^0. \quad (35)$$

The closed-form counterpart is

$$\mathbb{E}[(C_{c,i,t} - \hat{C}_{c,i,t})^2] \geq \left[\frac{2^{2H_{c,i,t}^0}}{2\pi e} - \frac{1}{12} \right]_+. \quad (36)$$

If Q_C top-codes the target, deterministic processing gives

$$H(Q_C(C) \mid \mathcal{F}) \leq H(C \mid \mathcal{F}). \quad (37)$$

For the bicycle-augmented information set,

$$H_{c,i,t}^B = H(C_{c,i,t} \mid \mathcal{F}_{c,i,t}^B), \quad (38)$$

$$L_{c,i,t}^B = \mathcal{H}_{\mathbb{Z}}^{-1}[H_{c,i,t}^B], \quad (39)$$

with the analogous closed-form bound

$$\mathbb{E}[(C_{c,i,t} - \hat{C}_{c,i,t}^B)^2] \geq \left[\frac{2^{2H_{c,i,t}^B}}{2\pi e} - \frac{1}{12} \right]_+. \quad (40)$$

Proposition 2 (Conditional information value of bicycle mobility). *The conditional entropy reduction is*

$$\Delta H_{c,i,t}^B = H_{c,i,t}^0 - H_{c,i,t}^B = I(C_{c,i,t}; Q_B(\mathbf{B}_{c,i,t}^{(q)}) \mid \mathcal{F}_{c,i,t}^0) \geq 0, \quad (41)$$

so

$$L_{c,i,t}^B \leq L_{c,i,t}^0. \quad (42)$$

For the discrete entropy-power shorthand

$$N_d(C \mid \mathcal{F}) = 2^{H(C \mid \mathcal{F})}, \quad (43)$$

$$\frac{N_d(C \mid \mathcal{F}^B)}{N_d(C \mid \mathcal{F}^0)} = 2^{-2\Delta H^B}. \quad (44)$$

Finally,

$$I(C; Q_B(\mathbf{B}) \mid \mathcal{F}^0) \leq I(C; \mathbf{B} \mid \mathcal{F}^0) \quad (45)$$

formalizes the information loss from binning or feature compression.

4.9. Entropy, CMI, and DELB Estimation

For a discrete target C and state S , conditional entropy was estimated from the empirical joint probability mass function. The Miller–Madow correction was the prespecified

primary finite-sample adjustment [60]; plug-in estimates were retained as diagnostic outputs. At city level,

$$\hat{I}_c^B = \hat{H}(C | S^0) - \hat{H}(C | S^B). \quad (46)$$

Lattice-envelope lower bounds were evaluated with the numerical inverse $\mathcal{H}_{\mathbb{Z}}^{-1}$ defined in Equation (7):

$$\hat{L}_c^0 = \mathcal{H}_{\mathbb{Z}}^{-1}[\hat{H}(C | S^0)], \quad \hat{L}_c^B = \mathcal{H}_{\mathbb{Z}}^{-1}[\hat{H}(C | S^B)]. \quad (47)$$

The quantities of interest were $\Delta \hat{L}_c = \hat{L}_c^0 - \hat{L}_c^B$ and $\hat{R}_c^L = \Delta \hat{L}_c / \hat{L}_c^0$. At grid level, the corresponding population definitions are

$$\Delta L_{c,i} = L_{c,i}^0 - L_{c,i}^B, \quad (48)$$

$$R_{c,i}^L = \frac{L_{c,i}^0 - L_{c,i}^B}{L_{c,i}^0}, \quad L_{c,i}^0 > 0. \quad (49)$$

When sampling fluctuation made an estimated CMI negative, only its contribution to the ordered theoretical bound was projected to zero; raw estimates were retained in diagnostic tables. No such projection was required for the pooled city estimates reported in the Results.

Before empirical estimation, we checked the lattice partition function, entropy-moment inversion, the zero-entropy boundary, and ordering of exact and closed-form bounds. The validation grid covered 0–8 bits, used a lattice-tail tolerance of 10^{-15} , and reproduced target discrete-Gaussian moments and entropies in simulations of up to 10^5 draws.

4.10. Sparse-State Bias and Randomization Calibration

We distinguish the population estimand, its finite-sample estimate, and the randomization reference. First, the population estimand is $\theta = I(C; B | S^0)$, which determines the ordering of the population DELB and Fano floors after side information is added. Its finite-sample estimate can be represented schematically as

$$\hat{\theta} = \theta + b(N, K_{\text{obs}}, S_1, \nu) + \varepsilon, \quad (50)$$

where K_{obs} is the number of observed joint atoms, S_1 is the share of observations in singleton atoms, ν is an effective conditional degrees of freedom, and ε denotes sampling variation. Equation (50) is an organizing representation, not a claim that the bias is known from these summaries alone.

The remaining object is the design-conditioned randomization distribution, $\mathcal{L}_{\pi}(\hat{\theta}^{\pi} | H_0)$. Because the estimator and state occupancy are recomputed after each transformation π , its mean need not equal zero. The observed statistic is compared with this reference distribution to assess evidence under the chosen transformation, not to validate the DELB or Fano inequalities. Figure 3 summarizes the distinction.

4.10.1. Known-Truth Sparse Nested-State Simulation

Finite-sample behavior was evaluated under a fully enumerated probability model. The baseline state Z had 4, 16, or 64 categories, the added state B had 2, 4, or 8 categories, and the integer target followed a negative-binomial conditional distribution. The factorial design crossed sample sizes 365 and 1095; balanced and skewed state probabilities; zero, moderate, and stronger population CMI; and independent or persistent Markov state sequences. Exact summation, with the truncated tail below the prespecified tolerance, supplied population CMI rather than treating a large simulated sample as truth.

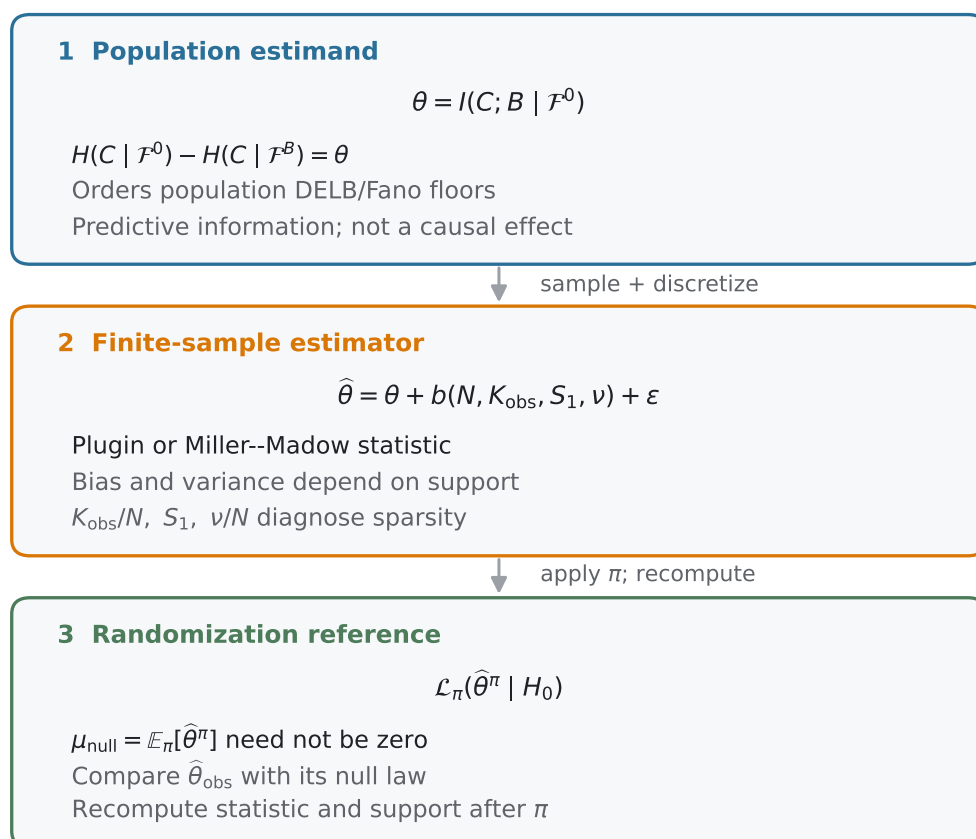


Figure 3. Population estimand, finite-sample estimator, and transformation-specific randomization reference. CMI denotes predictive information and is not interpreted as a causal effect.

For each of 216 estimator cells, 150 samples were generated, yielding 32,400 replicate records. Both plug-in and Miller–Madow CMI were computed with the same functions used in the empirical analysis, along with K_{obs}/N , S_1 , and ν/N . A second experiment applied the same temporal-block, spatial-series, and circular-shift transformations to population-null and positive-CMI samples. It comprised 30 Monte Carlo datasets per cell and 99 transformations per estimator and null design. The reported Type I rejection rate and power assess those specific data-generating and transformation mechanisms rather than all CMI estimators or randomization schemes.

4.11. DELB-Based Evaluation Algorithm

Algorithm 1 (DELB-based evaluation of an added urban data source). **Input:** integer target C ; baseline state S^0 ; added-data state B ; training, validation, and test indices; entropy estimator; randomization designs; number of repetitions R .

Output: $\hat{L}^0$, $\hat{L}^B$, $\hat{\Delta L}$, $\hat{I}(C; B \mid S^0)$, support diagnostics, randomization p -values, and matched held-out prediction errors.

1. Estimate all discretization thresholds from the training period and freeze them for validation and testing.
2. Construct the baseline state S^0 and augmented state $S^B = (S^0, B)$ on identical eligible observations.
3. Estimate $\hat{H}(C \mid S^0)$ and $\hat{H}(C \mid S^B)$ with the same entropy estimator.
4. Numerically evaluate $\hat{L}^j = \mathcal{H}_{\mathbb{Z}}^{-1}[\hat{H}(C \mid S^j)]$, $j \in \{0, B\}$.

5. Compute raw CMI, the absolute and relative estimated DELB reductions, and the support-density, singleton-share, and effective degrees-of-freedom diagnostics.
6. For each prespecified randomization design and repetition, transform B , reconstruct S^B , and recompute both state support and the complete statistic; then calculate plus-one upper-tail p -values.
7. Fit each prediction model twice with matched samples, training windows, tuning rules, and all non-bicycle inputs: once with S^0 and once with S^B .
8. Compare the estimated DELB change, randomization-calibrated evidence, and held-out MSE change as three separate outputs.

Numerical inversion.

For a positive entropy value, the implementation brackets the discrete-Gaussian rate λ from an initial value of 1 by repeated halving or doubling and then applies Brent's method. The root tolerance is 10^{-12} , with at most 200 iterations. For $\lambda \geq 1$, the partition function and second moment are evaluated by direct symmetric integer-lattice summation. For $\lambda < 1$, the Poisson-dual theta series avoids an unnecessarily wide real-lattice truncation. In both cases the support radius is expanded until the analytic bound on the omitted two-sided unnormalized tail is at most 10^{-15} . Entropy values at or below zero map to a DELB of zero. Repeated resampling and randomization paths use the configured monotone interpolation grid over 0–8 bits, with direct inverse evaluations retained for validation.

Computational cost and parallelization.

Let N be the number of matched observations, K_0 and K_B the numbers of occupied atoms in the two state tables, M the maximum retained half-support of a lattice sum, $T \leq 200$ the number of root evaluations, and J the number of entropy values converted to bounds. State counting requires $O(N)$ time and $O(K_0 + K_B)$ memory; direct bound conversion requires $O(JTM)$ time and $O(M)$ additional memory. With R randomizations, the serial calibration cost is $O(RN + RJTM)$, excluding model fitting, whose cost depends on the selected predictor. Randomizations are independent conditional on their stored seeds. The implementation evaluates city–spatial-scale–temporal-scale cells independently and records cell-level results and seeds for reproducibility.

4.12. City and Local DELB Estimation

City-level DELB point estimates were evaluated for both training-period-frozen analysis populations and separately by year. The deployment population uses the complete crime-supported domain, whereas the bicycle-covered population uses the domain in which the added mobility channel is observed. Both sets of cells were determined from 2020–2021 records and then applied unchanged to validation and testing; no 2022 outcome was used to redefine either spatial population. Their comparison therefore concerns the target analysis population rather than an administratively fixed city boundary.

Uncertainty was evaluated with 1000 bootstrap replicates. Each replicate was drawn by multinomial reweighting from the same fixed partition of the study calendar into non-overlapping seven-day blocks; the 1000 replicates were not themselves mutually disjoint. Resampling an entire city-day block preserved contemporaneous dependence among grids. Reported 95% intervals were centered bootstrap intervals. The paired decision-population comparison is limited to city-level point estimates. Local estimation, randomization calibration, held-out prediction, and multiscale analyses retain their prespecified bicycle-covered domains because complete-domain counterparts were not generated for all of those experiments.

The calculation was repeated within each grid after removing the grid identifier from the local state. The bicycle-covered spatial domain and bicycle-state thresholds were

frozen from 2020–2021, but the local-analysis support screen was descriptive and used the complete 2020–2022 series. It required 1095 valid lagged days, at least 30 crimes, at least 55 positive-flow days, no zero-flow run longer than 365 days, and at least two observed target and bicycle states. Thus, 2022 observations entered the support screen that produced the 339 local cells. These cells were used only for full-period local information mapping and randomization diagnostics; they were not used to select the grids or tune the models in the July–December 2022 held-out prediction experiment. We therefore do not interpret the local maps as leakage-free prospective validation. A one-sided normal screen based on the seven-day block bootstrap was adjusted by the Benjamini–Hochberg procedure [61]; because state-space sparsity can bias entropy differences, it is described as a *sampling-uncertainty screen*, not as a randomization test of conditional independence. Spatial clustering was summarized with global and local Moran statistics [62] using row-standardized queen contiguity and 999 permutations.

4.13. Three-Axis Scale and Sample-Size Design

To separate target changes from estimator stability, the design crossed 500 m, 1 km, and 2 km grids; daily and complete-week counts; and 20, 40, 60, 80, and 100% time-block samples on a nested common footprint. Stability was assessed against the full-sample estimate over 50 frozen thinning paths, not against unknown population truth. Count-level DELBs remained primary; intensity rescaling was applied only to the correspondingly rescaled count target. Daily, weekly, and spatially aggregated results are different estimands, and no DELB or CMI was divided by sample size. Randomization calibration covered all 90 city–space–time–sample cells with 199 temporal-block, spatial-series, and circular-shift replicates per cell; the three prespecified key scales used 999. Plus-one upper-tail p -values were adjusted across 270 tests and within city–null-design families. These comparisons are design-conditioned diagnostics, not exact conditional randomization tests. The nested-block sampling, normalization, support recomputation, and expanding-window details are available with the secondary analysis protocols.

Truth-referenced identifiability design.

At 500 m–day, 1 km–day, and 2 km–week, fixed-design semi-synthetic populations crossed $N \in \{90, 150, 365, 730\}$ generated periods with true CMI of 0, 0.03, 0.06, or 0.12 bits. Each of the 48 design cells contained 200 replicates evaluated by plug-in, Miller–Madow, and Jeffreys estimators. Power used baseline-state-matched surrogates (199 routinely; 999 for 12 prespecified cells). The 80% minimum detectable CMI was read from the tested effect grid, with interpolation reported separately. These conditional fixed-design populations are not recursively coherent crime processes; at 2 km–week, $N = 730$ refers to generated periods after deterministic state replay, not observed weeks. The full generation, support, and precision protocol accompanies the secondary analysis results.

4.14. Matched Prediction and Predictability Gap

Four model families were compared under the paired states in Equations (29) and (30): a smoothed state mean, a Poisson generalized linear model (GLM), a negative-binomial GLM, and histogram gradient boosting with Poisson loss. Within each family, baseline and bicycle-augmented specifications used the same target, sample, preprocessing, training period, tuning rule, and integer post-processing. The bicycle-augmented version added only $Q_B(B_{t-1}^{\text{total}})$. A seven-day seasonal naive predictor was retained as an external descriptive benchmark but was not matched to the DELB state and was not used in the principal comparisons.

Hyperparameters were selected by validation-period integer MSE. Continuous non-negative predictions were converted to integer counts using half-up rounding. The primary metric was

$$\text{MSE}_{c,k}^j = \frac{1}{N_c} \sum_{(i,t) \in \mathcal{T}_c} \left(C_{c,i,t} - \hat{C}_{c,i,t,k}^j \right)^2, \quad j \in \{0, B\}, \quad (51)$$

where k indexes the model family and $\mathcal{T}_c$ is the fixed July–December 2022 test set. Positive $\Delta \text{MSE}_{c,k} = \text{MSE}_{c,k}^0 - \text{MSE}_{c,k}^B$ denotes improvement. **One thousand bootstrap replicates drawn from a shared fixed partition into non-overlapping seven-day test blocks preserved cross-grid dependence.** Family-level screens used one-sided normal approximations and Benjamini–Hochberg adjustment.

4.15. Secondary Fano, Heterogeneity, and Robustness Designs

Fixed finite-label targets complemented the integer-count MSE analysis. The primary target was

$$Y_{c,i,t}^{(2)} = \mathbf{1}\{C_{c,i,t} > 0\},$$

and the robustness target mapped $C = 0$, $C = 1$, and $C \geq 2$ to three classes. These thresholds were fixed globally and were not selected from city or test outcomes. Baseline and bicycle-augmented states were exactly those in Equations (29) and (30).

Conditional entropy and the Fano floor were estimated on the combined training and validation period. **A statewise modal classifier assigns to each test observation the most frequent training-and-validation class within its observed conditioning state.** It was then evaluated on the unchanged July–December 2022 test period. Test states unseen before July 2022 used the pretest global mode; no test label was used to set a fallback. **Uncertainty used 300 bootstrap replicates drawn from a fixed partition into non-overlapping seven-day calendar blocks.** Centered intervals were reported because unconditional resampling can itself remove rare states; the raw bootstrap means were retained as diagnostics.

An oracle check reused the enumerated simulation models and compared the true Bayes error with the true Fano floor for binary and three-level targets. The empirical comparison contrasted an estimated pretest floor with a held-out classifier error. Estimated violations, if present, were retained as finite-sample or distribution-shift diagnostics. DELB and Fano values were never numerically compared because their targets and losses differ.

4.15.1. Predictability Gap and Theory–Practice Comparison

The model-specific predictability gap was

$$G_{c,k}^j = \text{MSE}_{c,k}^j - L_c^j \quad (52)$$

and its narrowing,

$$\Delta G_{c,k} = G_{c,k}^0 - G_{c,k}^B = \Delta \text{MSE}_{c,k} - \Delta L_c. \quad (53)$$

In population notation, the same accounting quantity is

$$G = \text{MSE} - L, \quad (54)$$

and an information-use ratio may be reported only when its denominator is positive:

$$\eta = \frac{\Delta \text{MSE}}{\Delta L}. \quad (55)$$

A positive ΔG means that empirical MSE fell by more than the estimated DELB, thereby narrowing the distance between the model error and the lower bound. A negative ΔG means that the DELB fell by more than the model MSE, so the added estimated prediction

opportunity was not fully realized; $\Delta G = 0$ means that they changed by the same amount. For the test comparison, the DELB values were re-estimated on the matched test-window information states.

For each held-out integer predictor, we additionally decomposed the entropy-domain nonattainment slack. Let $E = C - \hat{C}$, F denote the matched forecast-time state, and q_D be the discrete Gaussian with the same second moment $D = \mathbb{E}[E^2]$. We estimated

$$H(q_D) - H(E | F) = I(E; F) + D_{\text{KL}}(P_E \| q_D),$$

where the first term measures residual dependence on the forecast-time state and the second measures marginal error-shape mismatch. All probabilities were estimated on the same held-out empirical distribution. The decomposition is descriptive and finite-sample; it is not a causal attribution.

At the local scale, pretest $\Delta L_{c,i}$ from training plus validation was compared with test-period $\Delta \text{MSE}_{c,i,k}$. Spearman correlations used 2000 paired-grid bootstrap resamples and false-discovery-rate adjustment. Separate ordinary least-squares models regressed local empirical improvement on local theoretical reduction with city fixed effects and HC3 standard errors. These analyses are descriptive: cross-grid dependence was not modeled in the regression covariance.

4.15.2. Crime-Type Heterogeneity

The information and matched-prediction analyses were repeated for property theft, vehicle theft, and burglary, using category-specific lagged counts but the same bicycle thresholds and temporal split. This produced 18 city–category theory estimates (pooled and test), 36 paired test prediction contrasts, and paired within-city category comparisons. Seven-day block resampling and Benjamini–Hochberg adjustment were applied as in the primary experiments.

4.15.3. Alternative Specifications and Null Calibration

Prespecified robustness analyses examined whether the sign of the estimated DELB reduction persisted across design changes. The 16 theory specifications comprised 500 m, 1 km, and 2 km grids; daily and weekly targets; alternative crime and bicycle state mappings; crime and bicycle lags through seven days; and inflow, outflow, and absolute net flow. Each specification was estimated with plug-in, Miller–Madow, and a coherent Jeffreys–Dirichlet estimator. For the latter, a single symmetric Dirichlet posterior was placed on the observed atoms of (C_t, S_t^0, B_{t-1}) , and both conditional entropies were obtained by marginal aggregation of that posterior. Bootstrap intervals were recomputed with 7-, 14-, and 28-day blocks. Predictive stability was assessed at 12 expanding monthly origins in 2022, each with a one-month evaluation window and the same four model families.

Three diagnostics assessed whether the observed CMI exceeded finite-sample reference distributions. A temporal-block permutation reassigned seven-day mobility blocks, a spatial-series permutation reassigned entire mobility histories among grids, and a circular-shift surrogate displaced each grid’s mobility series by an admissible random offset. Each design used 999 transformations, upper-tail randomization p -values, and Benjamini–Hochberg correction. The same designs were applied to the 339 eligible grids. A local cell was declared null-calibrated only if it passed all three adjusted tests. Finally, matched 14- and 30-day mobility lags were compared with the recent one-day lag. These procedures are diagnostic randomization tests, not exact conditional randomization tests and not tests of a causal bicycle effect.

Estimator-sensitivity analysis assessed whether the city-level calibration depended on the primary Miller–Madow correction. On each of the same prespecified surrogates, it

recomputed plug-in, Miller–Madow, and coherent observed-support Jeffreys–Dirichlet CMI. Plus-one upper-tail p -values were adjusted in two prespecified families: nine city–design comparisons within each estimator and all 27 estimator–city–design comparisons globally. This analysis did not rerun the 339-cell local screen. The Jeffreys calculation regularizes the probabilities of observed joint states; it does not impute unobserved support.

A baseline-state-matched restricted randomization was also evaluated at the city level. Each grid series was partitioned into complete seven-day blocks. Blocks were permuted without replacement within grid, year, and season after ordering them by the distribution of lagged-crime states and holiday occupancy; strata with fewer than five blocks were widened within year. Terminal incomplete blocks remained fixed. The design used 999 randomizations, plus-one upper-tail p -values, and a three-city Benjamini–Hochberg family. Prespecified diagnostics required at least 80% of blocks to be movable and an absolute standardized difference no larger than 0.10 for the bicycle-state marginal.

A semi-synthetic experiment evaluated calibration and power on fixed empirical state supports. Six flow-quantile-spanning grids and 365 days were selected deterministically per city. Negative-binomial targets were generated from shrinkage estimates depending only on the baseline state under the null; centered bicycle-state effects were then calibrated by enumeration to target population CMI values of 0, 0.02, 0.05, and 0.10 bit, subject to the maximum attainable value on each support. The factorial design crossed three cities, fitted and enhanced-overdispersion scenarios, and the original temporal-block and restricted matched-block references. Each cell used 50 Monte Carlo datasets and 99 randomizations. Exact binomial 95% intervals accompanied Type I error and power. This restricted design is closer to the baseline state than the temporal-block, spatial-series, and circular-shift designs, but it is not an exact conditional randomization test because it does not sample from a known $B \mid \mathcal{S}^0$ distribution.

4.15.4. Surrogate-Specific Support Diagnostics

Each accepted surrogate was reconstructed from its original seed, after which both CMI and state support were recomputed. For each observed or randomized sample, the recorded quantities were

$$K_{\text{obs}} = \#\{(C, Z, B) \text{ observed atoms}\}, \quad S_1 = \frac{\#\{\text{observations in singleton atoms}\}}{N}, \quad (56)$$

and

$$\nu = \sum_z (r_z - 1)(c_z - 1), \quad (57)$$

where r_z and c_z are the observed target and bicycle-state cardinalities within baseline state z . The analysis used K_{obs}/N , S_1 , and ν/N .

Simulation-based associations related null bias to these support metrics. Empirical associations were reported descriptively with Spearman correlations and city and null-design fixed-effects regressions using city-grid-clustered standard errors. Models related the null mean to mean surrogate support and observed-minus-null CMI to surrogate-minus-observed support change. These associations diagnose a finite-support mechanism; they do not turn the null mean into an estimator-bias estimate or identify a causal effect.

4.16. Reproducibility

Analyses were specified in version-controlled configuration files and Python scripts. Random seeds, design parameters, sample indices, machine-readable results, and the inputs used to generate each table and figure were archived; manuscript graphics were generated programmatically and were not manually edited. A minimal known-truth example at `examples/minimal_delb_cmi_simulation.py` reproduces the population CMI ordering

and discrete-envelope DELB inversion without city data as a pedagogical complement to the full simulation design. Experiment lineage, output-acceptance checks, hash verification, descriptive figures, and secondary empirical analyses are documented together in the Supplementary Materials rather than through repeated interruptions in the main Methods.

5. Results

We first report the city-level DELB estimates and then examine finite-sample bias, randomization calibration, and held-out prediction. Multiscale, detectability, Fano, heterogeneity, and rolling analyses are presented as supporting checks.

5.1. Multicity Application Results at a Glance

Table 4. Application-level synthesis across the three cities. The transformation-reference column reports the smallest unadjusted *p*-value across the temporal-block, spatial-series, and circular-shift references; the matched-reference column reports the *p*-value from baseline-state-matched restricted randomization. Positive prediction screens are adjusted within the matched city–model comparisons.

City	Complete-domain ΔL	Covered ΔL	Covered relative reduction	Transform. min. <i>p</i>	Matched <i>p</i>	Positive screens / gap narrowing
Washington, DC	0.0119	0.0123	7.9%	0.967	0.599	0/4 / 0/4
New York City	0.0094	0.0604	17.3%	0.986	0.999	2/4 / 2/4
Vancouver	0.0115	0.0919	18.0%	0.903	0.994	1/4 / 0/4

Adding lagged bicycle flow lowered the DELB point estimate in all three cities, but neither the randomization analyses nor the prediction models showed a uniform gain (Table 4). In the complete crime-supported deployment populations, adding lagged bicycle flow yielded positive but small estimated DELB reductions. In the bicycle-covered populations, the corresponding relative reductions were 7.9% in Washington, DC, 17.3% in New York City, and 18.0% in Vancouver. The contrast was largest in New York City and Vancouver.

Neither the three transformation references nor the baseline-state-matched reference separated the observed CMI in any city under the primary grid-identity specification. A lower-cardinality sensitivity detected only a small New York difference after multiplicity adjustment. Held-out prediction was also selective: multiplicity-adjusted improvement tests were positive for two of four New York City models and one Vancouver model, whereas only the two New York City models narrowed the estimated Predictability Gap. Thus, lower DELB point estimates coexisted with non-rejection in the randomization analyses and heterogeneous held-out performance.

5.2. Estimated DELB Changes

For the bicycle-covered population used by the detailed calibration workflow, adding the one-day-lagged bicycle state reduced estimated conditional entropy in all three cities. The Miller-Madow CMI estimates were 0.043 bits in Washington, DC, 0.131 bits in New York City, and 0.142 bits in Vancouver. Mapping the baseline and bicycle-augmented conditional entropies through the same inverse integer-lattice envelope produced estimated DELB reductions of 0.012 MSE units in Washington, DC, 0.060 in New York City, and 0.092 in Vancouver. These changes represented relative reductions of 7.9%, 17.3%, and 18.0%, respectively (Figure 4). The centered seven-day block-bootstrap intervals for the CMI estimates excluded zero. These intervals quantify sampling variation under the observed state partition and dependence structure; randomization-based attribution is examined separately in Section 5.4.

The two training-period-frozen analysis populations retained positive city-level point estimates. For the deployment population, CMI was 0.0420, 0.0348, and 0.0377 bits in

Washington, DC, New York City, and Vancouver, respectively, compared with 0.0432, 0.1314, and 0.1423 bits for the bicycle-covered population. The corresponding deployment-population DELB reductions were 0.0119, 0.0094, and 0.0115 MSE units, versus 0.0123, 0.0604, and 0.0919 in the bicycle-covered populations. **The covered shares were 0.973, 0.246, and 0.241. Multiplying covered-domain CMI by these shares reproduced the Washington deployment estimate and yielded 0.0323 and 0.0343 bits for New York and Vancouver, leaving residuals of 0.0025 and 0.0034 bits. Because grid identity is in the conditioning state, the complete-domain CMI decomposes exactly into the observation-share-weighted CMIs inside and outside the frozen footprint. Direct estimation gives outside-footprint CMI of 0.00000, 0.00333, and 0.00445 bits in Washington, New York, and Vancouver; multiplying by the outside shares yields residual contributions of 0.00000, 0.00251, and 0.00338 bits. These values account for the 8–10% arithmetic difference in the two low-coverage cities. Separately, 2022 bicycle activity occurred outside the frozen training footprint (212 New York grids and 11 Vancouver grids), consistent with nondegenerate outside-footprint states. This decomposition identifies the population contributing the residual, not a causal mechanism or a share attributable specifically to 2022 activity. Thus, the sign was stable, while the magnitude difference in New York City and Vancouver was primarily a coverage-weighting effect. This paired comparison concerns city-level point estimates only; the downstream inferential experiments retain their prespecified bicycle-covered populations. A third set of DELB differences, 0.01511, 0.03363, and 0.05095 MSE units, was re-estimated on the covered July–December 2022 test window only to pair the bound with held-out model error. It is not a third estimate for the full 2020–2022 population. Supplementary Table S11 places these three populations, periods, and uses side by side.**

Local estimates were positive across most eligible cells, with detectable spatial clustering in Washington, DC and Vancouver but not New York City. Map colors represent point-estimate magnitude rather than calibrated local evidence.

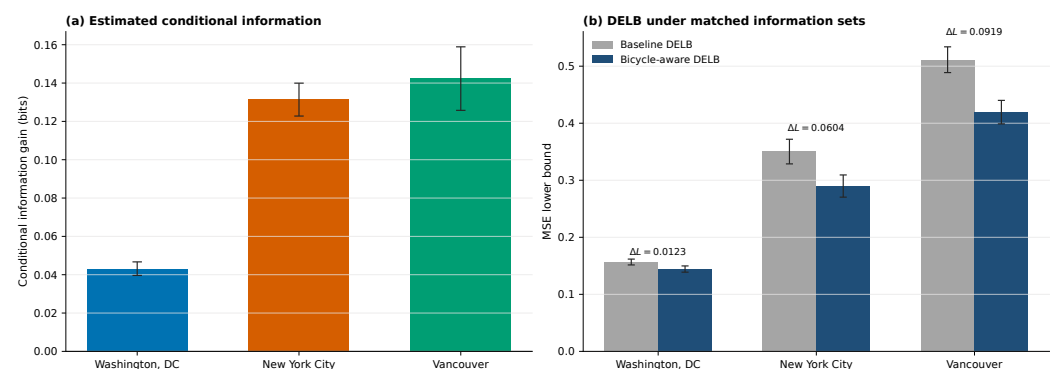


Figure 4. Bicycle-covered city-level point estimates. **(a)** Miller–Madow CMI with centered 95% seven-day block-bootstrap intervals. **(b)** DELB-implied MSE lower bounds under the baseline and bicycle-augmented information sets. These intervals describe sampling variation under the observed setup; randomization calibration is reported in Section 5.4.

5.3. Finite-Sample Behavior of CMI Estimation

Known-truth simulations were used to determine how sparse nested states affect CMI estimation and transformation-based calibration. Population CMI was obtained by exact enumeration, so estimator bias could be separated from the center of the corresponding randomization distribution.

Under the 72 population-null design cells, both estimators produced positive CMI estimates despite the true CMI being zero. Mean null bias was 0.1651 bits for the plug-in estimator and 0.1264 bits for the Miller–Madow estimator. Increasing the sample size from

365 to 1095 observations reduced average bias, whereas larger joint state spaces, greater singleton occupancy, and larger effective conditional degrees of freedom increased it. The Miller–Madow correction reduced average bias but did not uniformly outperform the plug-in estimator in the sparsest designs.

The transformation-specific reference distributions were also positively centered. Their mean centers were 0.2069 bits for the plug-in estimator and 0.1376 bits for the Miller–Madow estimator. Despite these nonzero centers, the mean Type I rejection rates were 0.014 and 0.011, respectively, because the observed statistics were compared with their complete transformation-specific reference distributions rather than with zero. Mean power across the positive-CMI designs was 0.337 for the plug-in estimator and 0.447 for the Miller–Madow estimator (Table 5 and Figure 5).

Table 5. Finite-sample CMI simulation summary. Null bias and null centers are in bits. Size is the mean rejection rate under population CMI equal to zero; power averages the positive-CMI designs.

Estimator	Mean null bias	Mean null center	Size	Power
Plug-in	0.1651	0.2069	0.014	0.337
Miller–Madow	0.1264	0.1376	0.011	0.447

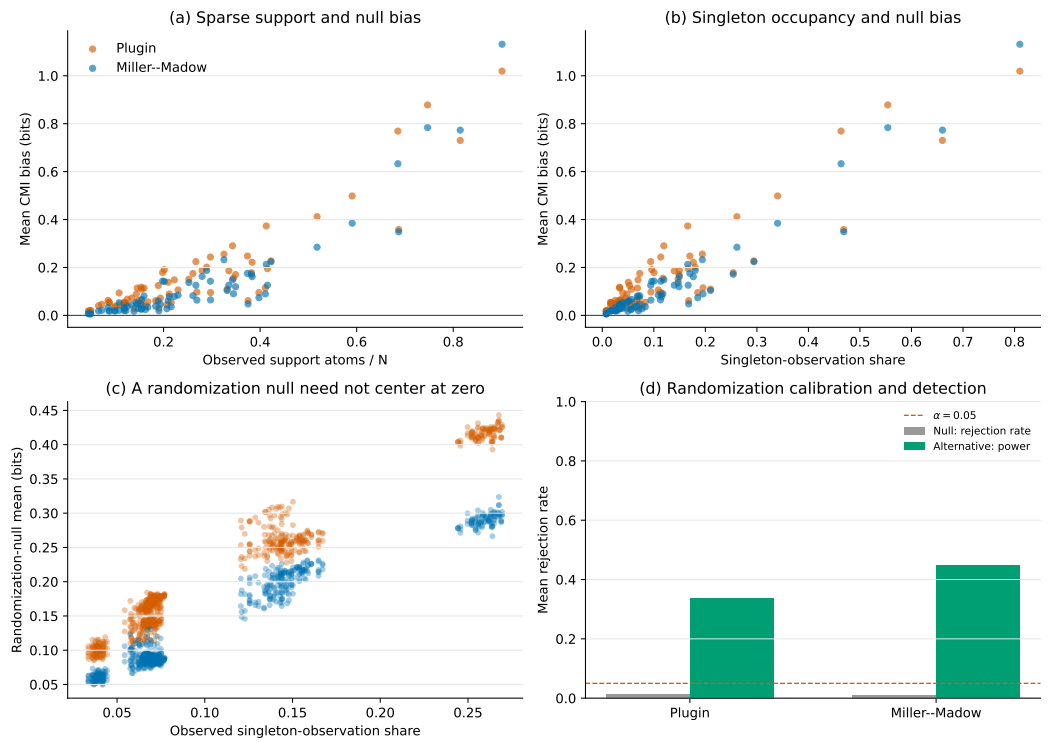


Figure 5. Known-truth simulation of sparse nested-state CMI estimation and randomization behavior. Population CMI is obtained by exact enumeration. Positive estimator bias and a positive randomization center are distinct finite-sample objects.

Support diagnostics showed consistent associations with finite-sample behavior. In the known-null simulations, observed-atom density, singleton-observation share, and normalized effective degrees of freedom were associated with the bias of both estimators. Across the 1017 empirical city–grid–reference summaries, their Spearman correlations with the randomization-reference mean were 0.835, 0.866, and 0.988, respectively. In a fixed-effects model with city-grid-clustered standard errors, surrogate normalized effective degrees of freedom had a coefficient of 0.7960 (SE 0.0257; $R^2 = 0.977$).

The transformations also changed the CMI support. Changes in support complexity between each surrogate and the observed table were negatively associated with observed-

minus-reference CMI, with correlations ranging from -0.653 to -0.791 . These relations are diagnostic rather than causal and do not define a universal bias correction. They show that the center of a finite-sample randomization distribution depends jointly on the estimator, observed support, and transformation design. Accordingly, the calibrated analyses below compare observed CMI with complete design-matched reference distributions rather than using zero as an automatic benchmark.

5.4. Randomization Calibration and Lag Specificity

The randomization analyses evaluated whether the observed CMI was unusually large relative to three prespecified transformation-based reference distributions. Evidence under a given design required the observed statistic to fall in the upper tail of that reference distribution; the reference center was not constrained to zero.

None of the nine city-by-reference comparisons rejected its corresponding upper-tail null. Unadjusted p -values ranged from 0.903 to 1.000, and all multiplicity-adjusted q -values were 1.000. In most comparisons, the observed CMI was below the corresponding reference mean. At the local level, none of the 339 eligible cells passed all three randomization designs, including cells that had passed the block-bootstrap sampling-uncertainty screen.

The estimator-sensitivity analysis produced the same city-level conclusion for all three entropy estimators. Across the nine comparisons within each estimator, raw p -values ranged from 0.980 to 1.000 for plug-in, 0.903 to 1.000 for Miller–Madow, and 0.984 to 1.000 for Jeffreys–Dirichlet. No comparison passed either the estimator-specific nine-test correction (0/9) or the global 27-test correction (0/27), consistently across estimator-specific null summaries.

The distant-lag analysis produced similar CMI estimates. Ratios of 14-day- or 30-day-lag CMI to the matched one-day-lag CMI ranged from 0.964 to 1.052. Both distant-lag estimates exceeded the one-day estimate in New York City, whereas differences in Washington, DC and Vancouver were small and inconsistent (Table 8).

The similarity of one-, 14-, and 30-day estimates is more consistent with slow-moving exposure, seasonality, or another persistent common driver than with information specific to bicycle activity on the preceding day. The complete local eligibility statistics, maps, and detailed lag figure are retained as secondary diagnostics.

The randomization and distant-lag comparisons did not isolate an information signal specific to recent, geographically matched bicycle activity, although this conclusion remains conditional on the transformations examined.

Table 6. Baseline-state-matched restricted randomization and semi-synthetic calibration. Type I error and power use 50 Monte Carlo datasets and 99 randomizations per dataset under the fitted-dispersion scenario.

City	Observed CMI	Null mean	p	Type I error	Power at 0.02 bit	Power at 0.05 bit	Power at highest evaluated CMI
Washington, DC	0.0432	0.0433	0.599	0.04	0.10	0.06	0.48 at 0.0649 bit
New York City	0.1314	0.1338	0.999	0.08	0.02	0.02	0.46 at 0.1000 bit
Vancouver	0.1423	0.1476	0.994	0.00	0.02	0.02	0.30 at 0.1000 bit

The baseline-state-matched restricted randomization reached a movable-block share of 1.00 in every city and preserved the bicycle-state marginal exactly. Its null means were again slightly above the observed CMI (Table 6). The upper-tail p -values were 0.599, 0.999, and 0.994 in Washington, DC, New York City, and Vancouver; all three BH-adjusted q -values were 0.999. Thus, bringing the reference closer to the baseline state did not reverse the non-rejection under the broader transformation designs.

Table 7 places this result beside observed support diagnostics and the lower-cardinality sensitivity. Replacing grid identity with training-only crime-intensity quintiles reduced K_{obs}/N from 0.236–0.412 to 0.032–0.180. Washington and Vancouver remained near their matched-reference centers. New York CMI was 0.03625 bits, 0.00105 bits above the 999-surrogate mean (plus-one $p = 0.001$, three-city BH $q = 0.003$). The magnitude is small and does not validate the larger primary point estimate; it shows that state complexity controls finite-sample resolution.

Table 7. Observed-support diagnostics and matched-reference calibration. The primary specification uses grid identity; the lower-cardinality sensitivity replaces it with training-only crime-intensity quintiles. The difference is observed CMI minus the matched-reference mean.

Specification	City	K_{obs}/N	S_1	ν/N	Observed CMI	Null mean	CMI-null (bits)	$p / \text{BH } q$
Primary	Washington, DC	0.236	0.103	0.061	0.04316	0.04327	-0.00011	0.599 / 0.999
Primary	New York City	0.359	0.195	0.131	0.13137	0.13381	-0.00244	0.999 / 0.999
Primary	Vancouver	0.412	0.223	0.186	0.14233	0.14757	-0.00524	0.994 / 0.999
Lower-cardinality	Washington, DC	0.032	0.008	0.026	0.01315	0.01291	0.00024	0.165 / 0.248
Lower-cardinality	New York City	0.057	0.014	0.050	0.03625	0.03520	0.00105	0.001 / 0.003
Lower-cardinality	Vancouver	0.180	0.067	0.135	0.07186	0.07178	0.00008	0.480 / 0.480

All information values are in bits. Primary results use 999 surrogates. Lower-cardinality routine screens use 199 surrogates, with 999 for the New York confirmation. S_1 is the share of observations in singleton joint atoms. A separate known-truth null simulation yielded a mean Miller–Madow bias of 0.1264 bits; it illustrates sparse-state bias but is not a city-specific bias correction.

Two higher-complexity sensitivities provided no calibrated separation. Replacing season with calendar month produced observed-minus-null differences of -0.00287, -0.01623, and -0.02482 bits; using zero plus positive-flow quintiles produced -0.00211, -0.00272, and -0.00864 bits in Washington, New York, and Vancouver, respectively.

The semi-synthetic results qualified that finding. Under zero population CMI, restricted-design rejection rates ranged from 0.00 to 0.08 across cities and dispersion settings, and every exact 95% binomial interval covered 0.05. However, power at 0.02 and 0.05 bit was only 0.00–0.10. Even under the fitted-dispersion scenario, power at the highest evaluated attainable CMI was 0.48 in Washington, DC, 0.46 in New York City, and 0.30 in Vancouver. Some requested high-CMI settings were unattainable on the fixed empirical support and were reported at their actual maxima. The semi-synthetic analysis therefore supports calibration of the restricted reference but also shows limited detection power; non-rejection cannot establish conditional irrelevance.

Table 8. Randomization-reference calibration and distant-lag diagnostics. Randomization p -values are upper-tail tests of whether observed CMI exceeds the specified null distribution. Distant-lag columns report $\hat{I}_{\text{distant}}/\hat{I}_{\text{recent}}$.

City	Randomization p -value			Distant/recent CMI	
	Temporal	Spatial	Circular	14 days	30 days
Washington, DC	1.000	1.000	0.967	0.992	0.985
New York City	1.000	0.986	0.993	1.020	1.052
Vancouver	1.000	0.969	0.903	0.964	1.007

A post hoc coordinate-source sensitivity showed that station-reference matching can affect the Washington, DC estimate. Removing the 6.3929% of in-domain bicycle flow contributed by reference-located endpoints, while holding the crime sample, grid, and analysis domain fixed, increased its estimated CMI from 0.043157 to 0.063098 bits and its DELB reduction from 0.012324 to 0.017857 MSE units. Temporal and spatial reference tests

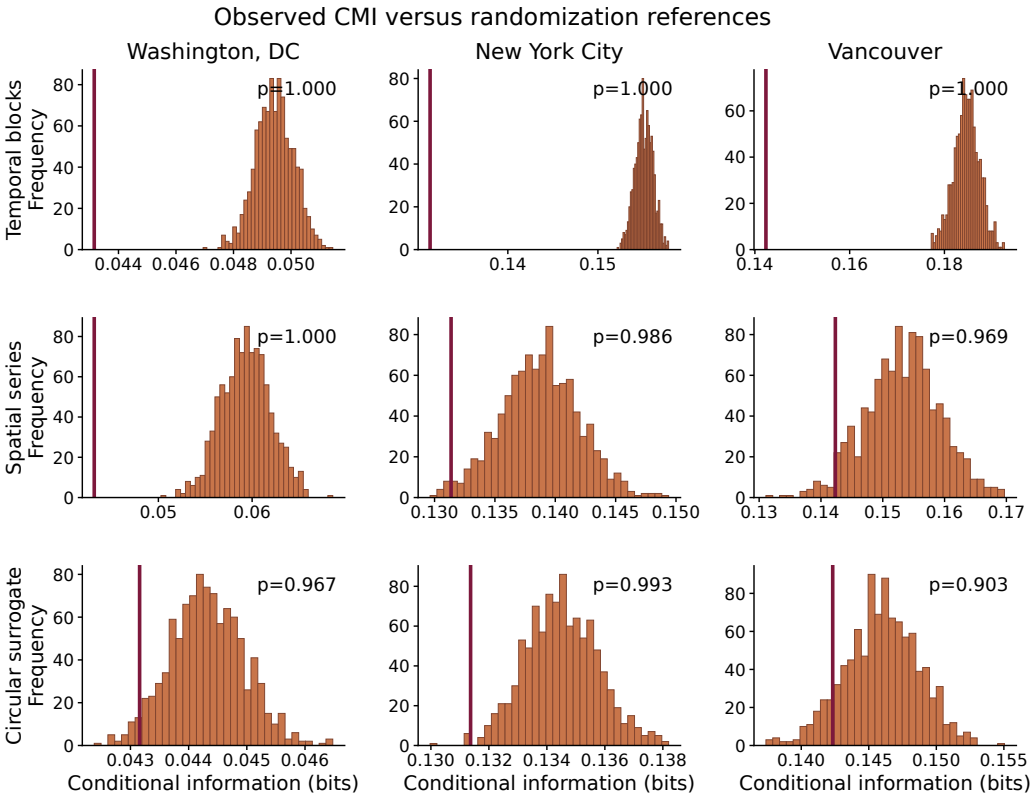


Figure 6. Observed city-level CMI relative to temporal-block, spatial-series, and circular-shift reference distributions. The observed statistic did not exceed any prespecified upper-tail reference. Burgundy lines mark observed CMI; terracotta bars show the reference distributions.

still did not separate the observed statistic (both $p = 1.000$ with 199 surrogates), but the circular-shift comparison yielded $p = 0.007$ in a 999-surrogate confirmation. In New York City only one in-domain flow contribution was removed, leaving the binned state and estimates unchanged. Vancouver has no source coordinates in its trip files and therefore no comparable source-only analysis. These exploratory results do not change the frozen primary randomization result or establish a robust three-city signal.

5.4.1. Spatiotemporal Scale and Sample-Size Calibration

The multiscale analysis separated target changes caused by aggregation from within-design stability and randomization calibration. All 18 full-sample city–space–time combinations yielded positive estimated DELB reductions, and estimates became more stable as the retained time-block fraction increased. Nevertheless, none of the 270 city–space–time–sample–reference tests survived global or within-family FDR correction (minimum raw $p = 0.005$; minimum within-family $q = 0.075$). Because aggregation changes the outcome and MSE unit, these results do not imply a scale-invariant effect.

The smallest isolated value was $p = 0.006$ for the New York City 500 m–day circular-shift comparison, but it did not survive the prespecified FDR correction. Across the 90 full-sample design cells, 21 circular-shift statistics exceeded their reference means compared with none under the temporal-block design. This asymmetry shows that the reference mechanisms remove different portions of temporal and spatial structure and should not be treated as interchangeable tests of one universal null.

5.4.2. Truth-Referenced Convergence and Detectability

The known-truth experiment showed that increasing the generated record length generally reduced CMI estimation error, but it did not produce a single stability threshold shared by all scales and estimators (Figure 7). Across positive-CMI cells and the three estimators, median CMI RMSE was 0.159 bits at $N = 90$, 0.167 at $N = 150$, 0.135 at $N = 365$, and 0.119 at $N = 730$. At the 2 km-week scale, the corresponding scale-specific median declined from 0.266 to 0.152 bits. DELB relative RMSE was less monotone: its overall median was 0.734, 0.666, 0.432, and 0.521 over the same four sample lengths. Thus, more generated periods improved CMI recovery overall, whereas nonlinear entropy-to-DELB inversion and sparse support prevented a universal convergence rule.

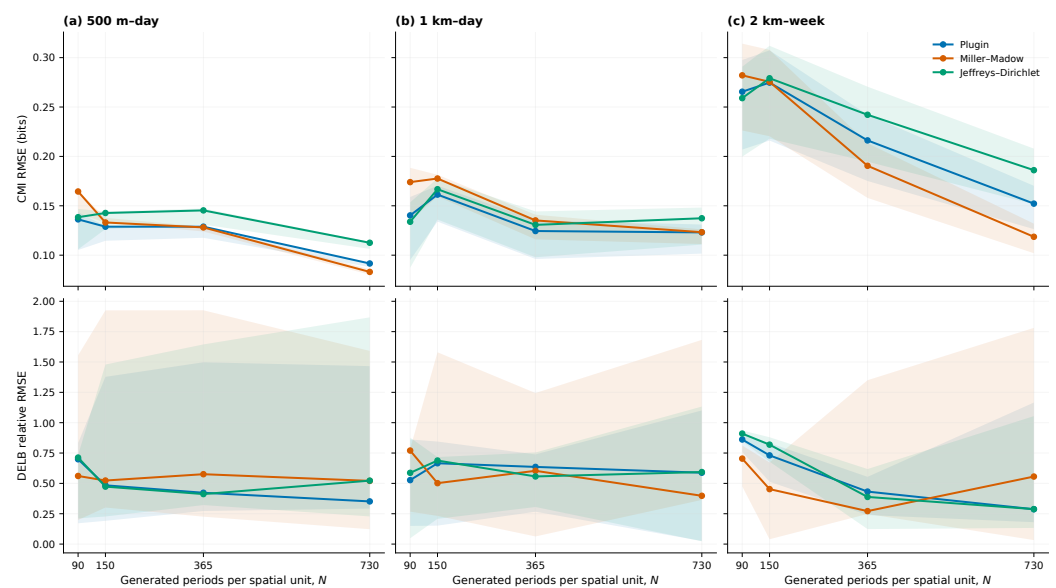


Figure 7. Truth-referenced CMI and DELB convergence in the semi-synthetic scale-sample experiment. Lines are medians over the three positive population-CMI levels; shaded bands span those levels. N denotes generated periods per spatial unit.

The exact baseline-state-matched reference was approximately calibrated under zero population CMI. Type I error ranged from 0.03 to 0.08 across the 12 scale-sample cells, with a median of 0.05. Power nevertheless varied substantially at small information gains. At 0.03 bit it ranged from 0.115 to 1.000 (median 0.703), at 0.06 bit from 0.340 to 1.000 (median 0.995), and at 0.12 bit from 0.885 to 1.000 (Figure 8).

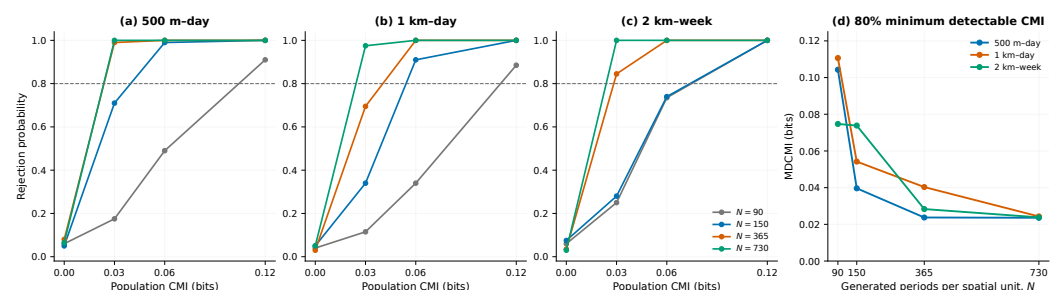


Figure 8. Power and 80% minimum detectable CMI for the exact baseline-state-matched reference. The dashed line marks 80% power. Interpolated MDCMI values use adjacent tested CMI levels and are not additional simulated effect levels.

The two power studies answer different questions. The earlier empirical-support experiment uses six selected grids per city and the fitted city supports to qualify the primary city non-rejections; it is therefore the direct governing power analysis for those

results. The scale-sample experiment varies record length and aggregation under known injected CMI to study design resolution. In the broader sparse-state simulation, Miller-Madow Type I rejection averaged 0.011 and plug-in rejection averaged 0.014, both below 0.05. This conservative behavior, together with low power at small CMI, weakens any interpretation of non-rejection as evidence of no information.

All 12 scale-sample combinations reached 80% power within the tested 0.12-bit grid. The interpolated 80% minimum detectable CMI ranged from 0.0236 to 0.1106 bits. At $N = 90$, it was 0.1043, 0.1106, and 0.0747 bits for 500 m-day, 1 km-day, and 2 km-week, respectively; at $N = 730$, the corresponding values were 0.0236, 0.0243, and 0.0238 bits. These values quantify design-specific detection limits and do not define a scale-free information threshold.

5.5. Held-Out Prediction and Predictability Gaps

Held-out prediction evaluated whether particular fitted models converted the augmented information state into lower error on unseen observations. The fixed July–December 2022 bicycle-covered test set contained 413 grids observed over 184 days, or 75,992 grid-day observations.

Adding the bicycle state reduced the MSE point estimate in five of the 12 city-model comparisons. Three comparisons passed the multiplicity-adjusted improvement test: the state-mean and Poisson gradient-boosting models in New York City and the state-mean model in Vancouver. The remaining model-city combinations showed either negligible change or higher MSE after the bicycle state was added (Figure 9).

Every baseline and bicycle-augmented held-out MSE remained above its corresponding DELB (Table 9). A positive gap narrowing means that the model's MSE decreased by more than the estimated DELB, so the model moved closer to the lower bound. By this definition, the New York City state-mean and gradient-boosting models narrowed the Predictability Gap by 0.001 and 0.029 MSE units, respectively. A negative value means that the DELB decreased by more than the model MSE, so the added estimated prediction opportunity was not fully realized by that model. This pattern occurred in all other comparisons.

For practical interpretation, consider the New York City baseline DELB of 0.263 and the bicycle-augmented DELB of 0.230. Under the corresponding estimated distributions, these values are lower bounds for admissible integer-valued predictors using the stated information sets; they are not forecasts of the MSE that a fitted model will automatically achieve. The observed model MSEs near 1.9 therefore indicate substantial nonattainment of the estimated floor. The 0.034-unit DELB reduction represents additional estimated prediction opportunity associated with the augmented information set. It becomes a realized forecasting gain only to the extent that a fitted model also reduces held-out MSE, which occurred selectively in Table 9.

The estimated DELB declined in every city, whereas held-out MSE improved in five of the 12 city-model comparisons and passed the multiplicity-adjusted improvement screen in three. Only the two New York City models narrowed the estimated Predictability Gap.

The held-out two-slack decomposition reconciled numerically in all 24 city-model-information-set cells (maximum absolute identity residual 3.30×10^{-13} bits). Residual error-state dependence ranged from 0.769 to 1.555 bits, whereas marginal error-shape mismatch ranged from 0.156 to 0.276 bits. Thus, most measured entropy-domain nonattainment was associated with residual dependence on the forecast-time state, not with the marginal error distribution alone. The components are finite-sample descriptive quantities; model-specific values for both information sets appear in Supplementary Materials, Table S13.

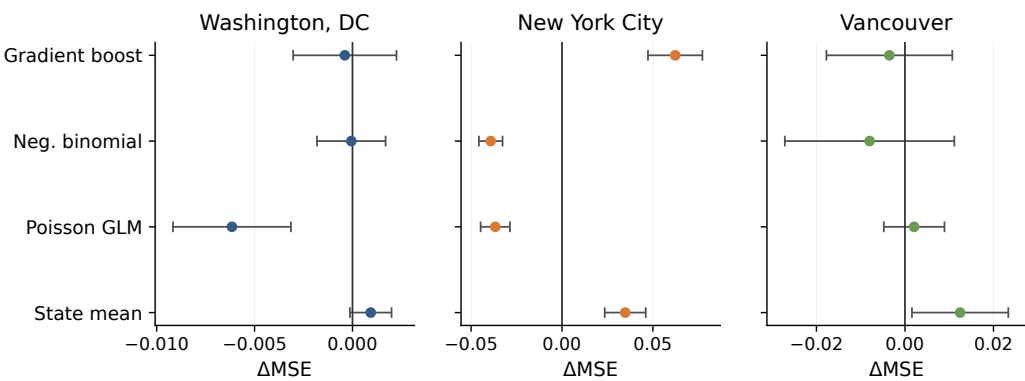


Figure 9. Held-out integer-MSE change after adding lagged bicycle state. Positive values favor the bicycle-augmented specification; comparisons are paired within city and model family.

Table 9. Matched held-out integer-prediction MSE, DELB, and Predictability Gap results for July–December 2022. The paired DELB column reports the baseline and bicycle-augmented lower bounds as L^0/L^B . Positive ΔMSE denotes lower error after adding bicycle information. Gap narrowing is $\Delta G = \Delta\text{MSE} - \Delta L$: a positive value indicates that the model moved closer to the lower bound, whereas a negative value indicates that the estimated lower bound decreased more than the model MSE. Differences were calculated from unrounded values and then rounded for display.

City	Model	MSE ⁰	MSE ^B	L^0/L^B	ΔMSE	ΔL	Gap narrowing
Washington, DC	State mean	0.5002	0.4993	0.107/0.092	0.0009	0.015	-0.014
Washington, DC	Poisson GLM	0.4933	0.4994	0.107/0.092	-0.0062	0.015	-0.021
Washington, DC	Negative-binomial GLM	0.4931	0.4932	0.107/0.092	-0.0001	0.015	-0.015
Washington, DC	Poisson gradient boosting	0.4952	0.4956	0.107/0.092	-0.0004	0.015	-0.016
New York City	State mean	2.0893	2.0545	0.263/0.230	0.0348	0.034	0.001
New York City	Poisson GLM	1.9043	1.9410	0.263/0.230	-0.0367	0.034	-0.070
New York City	Negative-binomial GLM	1.9073	1.9465	0.263/0.230	-0.0392	0.034	-0.073
New York City	Poisson gradient boosting	1.9010	1.8388	0.263/0.230	0.0623	0.034	0.029
Vancouver	State mean	1.6632	1.6507	0.259/0.208	0.0125	0.051	-0.038
Vancouver	Poisson GLM	1.6287	1.6266	0.259/0.208	0.0021	0.051	-0.049
Vancouver	Negative-binomial GLM	1.6138	1.6218	0.259/0.208	-0.0080	0.051	-0.059
Vancouver	Poisson gradient boosting	1.6654	1.6690	0.259/0.208	-0.0035	0.051	-0.054

5.6. Secondary and Robustness Analyses

5.6.1. Fano Classification-Loss Comparison

All 64 population-oracle comparisons satisfied Fano’s inequality, and all empirical held-out modal-classifier errors remained above their corresponding pretest Fano floors. No point-estimate violations were observed.

Adding the bicycle state lowered the estimated Miller–Madow Fano floor in all six city–target comparisons. Reductions ranged from 0.0104 to 0.0376. In contrast, held-out modal-classifier error increased in all six comparisons alongside more unseen bicycle-augmented test states.

Here, “unseen” refers to test observations whose complete conditioning state was absent before July 2022, not to a count of distinct states. For example, the New York entries 2,058/4,408 are baseline/bicycle-aware fallback counts out of 36,432 test observations (5.65%/12.10%).

The two movements share a finite-sample mechanism in this comparison: splitting a sparse baseline state by bicycle category lowers the estimated conditional entropy and hence the estimated Fano floor, while increasing the number of test observations assigned to an unseen augmented state and the pretest global-mode fallback. The simultaneous floor reduction and classifier degradation are therefore diagnostic of state refinement under sparse support; they do not establish a population information gain. The Fano and

DELB magnitudes are not directly comparable because they concern different targets and loss functions. Crime-type estimates were positive but category-specific prediction gains were sparse, and rolling-origin gains remained concentrated in selected New York City models. Alternative estimators and specifications preserved the sign of the DELB point estimate but did not replace randomization calibration. **Complete local, multiscale, Fano, crime-type, alternative-specification, descriptive, and rolling-origin results are grouped in Supplementary Materials, Sections S2–S6 and are treated as secondary or exploratory evidence.**

6. Discussion

6.1. What the DELB Measures

The DELB is an information-constrained MSE lower bound for non-anticipating integer-valued prediction under a specified target, information set, forecast horizon, and squared-error loss. It maps conditional Shannon entropy through the inverse integer-lattice maximum-entropy envelope and therefore provides a model-independent benchmark for every admissible integer-valued predictor. Theorem 1 states its relation to the infimum risk and the conditions for equality. The predictor-dependent refinement and two-component decomposition then separate residual error–history dependence from error-distribution shape mismatch.

6.2. Why Estimated Information Is Not Calibrated Evidence

At the population level, adding side information reduces conditional entropy by its CMI with the target given the baseline information, and monotonicity of the inverse lattice envelope orders the corresponding DELBs. Estimating that ordering from sparse nested states is a separate inferential problem. The known-truth simulations showed that positive plug-in and Miller–Madow CMI estimates can arise when population CMI is zero. Dividing each baseline state into weakly occupied joint states makes the estimate depend on sample size and support geometry; standard corrections reduce some of this behavior but do not remove it.

Estimator sensitivity produced the same substantive caution. Plug-in, Miller–Madow, and observed-support Jeffreys–Dirichlet estimators preserved positive DELB-reduction point estimates across the prespecified specifications, yet none separated the observed city-level CMI from its reference distribution. The Jeffreys result regularizes probabilities on observed support and does not reconstruct unobserved states.

A transformation-based reference must likewise be distinguished from population CMI and estimator bias. Each transformation changes the data structure, after which the statistic and its support are recomputed. Its reference center may therefore be positive. The support diagnostics showed that effective conditional degrees of freedom and transformation-induced support changes tracked this centering. These associations are diagnostic, not a universal bias formula. Valid interpretation remains specific to the estimator, sample, support, and transformation design.

These results lead to the following evidence chain:

$$\begin{aligned} \text{population DELB/CMI} &\longrightarrow \text{finite-sample estimate} \longrightarrow \text{estimator behavior} \\ &\longrightarrow \text{design-conditioned reference} \longrightarrow \text{realized prediction error.} \end{aligned} \quad (58)$$

A positive estimate describes the observed partition, a randomization comparison evaluates a specified reference mechanism, and held-out prediction evaluates a fitted model. The truth-referenced scale–sample experiment further showed that sample size changes

estimation error and detection probability while the population DELB remains fixed. It did not identify a universal sample-length or scale-free detectability threshold.

The lower-cardinality sensitivity makes the role of state design concrete. Reducing the spatial partition greatly improved occupancy and revealed a statistically distinguishable but very small New York difference. Washington and Vancouver remained centered on their matched references. The appropriate conclusion is therefore neither that all positive estimates are artifacts nor that the primary magnitudes are confirmed: finite-sample identifiability changes with the conditioning-state resolution.

6.3. Interpretation of the Bicycle-Mobility Results

The bicycle-augmented state yielded lower conditional-entropy and DELB point estimates in every city. The reductions were substantially smaller in the complete crime-supported deployment populations than in bicycle-covered areas, especially in New York City and Vancouver. Coverage is therefore an integral part of the target population, beyond its role in missing-data handling.

The primary grid-identity randomization analyses did not isolate a signal specific to recent, geographically matched bicycle activity: no primary city-level or multiscale comparison survived the relevant calibration, and distant lags produced similar estimates. The lower-cardinality sensitivity identified only the small New York difference reported above. Because the matched-reference experiments had limited power for small signals, non-rejection limits source-specific attribution but does not establish that bicycle mobility is conditionally irrelevant.

The coordinate-source sensitivity further limits a blanket reading of the Washington, DC primary null result. Excluding station-reference-located endpoints increased its estimated CMI and DELB reduction and produced circular-shift separation in a post hoc analysis, while temporal and spatial references remained non-separating. This dependence on endpoint provenance and reference design merits independent validation; it should not be interpreted as confirmation of a city-wide mobility effect. New York City was unchanged under the source-only restriction, and Vancouver lacks the source coordinates needed for that comparison.

Coverage explains most of the contrast between the bicycle-covered and deployment populations. New York and Vancouver covered only about one quarter of the crime-supported grid-days, and coverage-weighted CMI closely approximated the complete-domain estimate. The remaining differences are the observation-share-weighted CMIs outside the frozen footprint. Bicycle activity was observed there in 2022, but the decomposition cannot attribute a specific share of those differences to that year or identify a causal mechanism.

6.4. Prediction Opportunity versus Model Realization

The held-out results illustrate why a lower information-constrained bound and lower realized error are different outcomes. Added information may enlarge the theoretically available opportunity even when a finite-sample estimator cannot exploit it. Conversely, a model may improve by using seasonal or proxy structure without establishing bicycle-specific population CMI.

This pattern was visible across cities and model families. Estimated DELBs declined in every city, but held-out improvements were selective, and only two New York City models narrowed the Predictability Gap. Under classification loss, every estimated augmented Fano floor declined while the statewide modal classifier worsened as unseen augmented states became more frequent. Neither pattern violates its lower bound.

The Predictability Gap functions as a matched accounting quantity whose MSE and DELB components must use the same target, forecast horizon, information set, evaluation sample, and loss. A negative gap change means that the tested model did not realize the added estimated opportunity; it does not invalidate the DELB.

The empirical decomposition further shows that residual error–history dependence is larger than marginal error–shape mismatch for every evaluated model. Closing the observed gap would therefore primarily require predictors that better use the declared forecast-time state, although the decomposition does not identify which new model will do so.

6.5. Scope, Limitations, and Transferability

For urban analytics teams, the framework supports staged data-value assessment before procurement or deployment decisions based on one fitted model. Table 10 translates the evidence chain into four operational questions: whether the estimated floor changes, whether finite-sample support is adequate, whether the statistic separates from design-matched references, and whether matched models realize a held-out benefit.

Table 10. Practical decision framework for evaluating an added urban data source in an integer-count forecasting system.

Decision question	Required evidence	Operational response	Boundary
Does the source create an estimated prediction opportunity?	DELB and CMI point estimates in frozen target populations	Retain the source for calibrated evaluation if the estimated floor declines materially	Point estimates alone are insufficient for source attribution or realized benefit
Is the apparent gain distinguishable from finite-sample structure?	Support diagnostics, known-truth calibration, and design-matched randomization	Attribute the gain only to the tested design and only when calibration and power support separation	Low-powered non-rejection is inconclusive; randomization provides no causal identification
Can available models use the source on unseen data?	Matched held-out models with identical targets, samples, tuning rules, and loss	Deploy only city–model combinations showing stable held-out benefit; monitor the Predictability Gap	Model-specific gains require separate validation across cities and model families
Does value persist outside the service footprint?	Training-period-frozen coverage-conditioned and complete crime-supported deployment populations	Report both populations and treat coverage as part of the data product	Service-area estimates apply only to covered locations

Any subsequent deployment assessment should remain city-, model-, and population-specific, retain a baseline model and frozen holdout, monitor both Δ MSE and Δ G, and reassess benefits under rolling evaluation.

Several limitations bound this application. Bicycle trips provide a mode-specific activity trace with incomplete coverage of the ambient population; coverage, adoption, operations, weather, and recording differ by city. Crime records depend on reporting, classification, and jurisdiction, and 2020–2022 includes pandemic-related structural change. Aggregate grid–day associations do not support individual inference or a causal effect of bicycle activity on crime.

Comparable forecast-time weather fields were not available in the frozen harmonized panel, so weather and daylight were not controlled. Adding city-specific sources after inspecting results would weaken comparability; this omission is reported as a limitation rather than addressed by a partial-city analysis. Bicycle flow may therefore carry weather- or daylight-related information beyond the coarse calendar state. Since those variables can be obtained separately, the present estimates cannot establish that maintaining a bicycle

feed is preferable to using less costly environmental data for the same forecasting decision. Annual CMI varied across 2020–2022, especially in New York and Vancouver, so the pandemic-era estimates should not be treated as stationary or directly transferable to later operating regimes.

The randomization designs do not preserve the exact mobility distribution within every baseline state. The known-truth and power experiments cover finite fixed-design families, and the support regressions are descriptive rather than analytic corrections. Some fixed-support CMI targets were unattainable; matched-reference power was limited for small signals; and minimum-detectable-CMI interpolation is grid based. Spatial and temporal aggregation also changes the target distribution and MSE unit, so the multiscale results do not establish a universal scale law.

Future work should focus on three issues: partial pooling for sparse states, randomization designs that better preserve the conditional mobility distribution, and evaluation with broader mobility and environmental data. The workflow is also transferable to demand, epidemiological, traffic, and service-arrival counts when the target, information set, and calibration design are clearly specified.

These implications concern forecasting-system evaluation, not operational policing decisions. Any public-sector deployment requires separate fairness, privacy, governance, and causal-impact assessment.

In particular, an information-value estimate should not by itself justify procuring surveillance-oriented data, expanding data collection, or allocating police resources. Procurement decisions must separately consider necessity, proportionality, representational bias, community oversight, retention and access controls, and whether less intrusive data can answer the same planning question.

7. Conclusions

We developed a finite-sample evaluation procedure that links conditional Shannon entropy to an integer-valued MSE lower bound and assesses whether an estimated reduction survives support diagnostics, randomization calibration, and held-out evaluation.

In all three cities, adding lagged bicycle flow lowered the DELB point estimate. The reduction was substantially smaller over the complete crime-supported populations than within bicycle-covered areas, and the observed CMI was not unusually large under any prespecified randomization reference in the primary specification. A lower-cardinality sensitivity identified a small calibrated difference only in New York City, demonstrating that empirical resolution depends on state complexity. Held-out improvements were limited to selected city–model combinations. A post hoc exclusion of station-reference-located bicycle endpoints also changed the Washington, DC point estimate and circular-shift diagnostic, so that city’s inference is sensitive to coordinate provenance.

These results show that a lower estimated prediction bound does not by itself demonstrate source-specific information or practical forecasting value. Data-source evaluations should report the estimated bound change, randomization-based evidence, and held-out prediction gain as separate results.

Supplementary Materials: The independent Supplementary Materials file contains estimator sensitivity and the training-period-frozen decision-population comparison, local and lag-specificity details, sample-size and multiscale analyses, crime-type heterogeneity, rolling-origin results, and additional descriptive and finite-sample figures.

Author Contributions: Concept and design: M.D., Y.Z.; data collection: M.D.; analysis and interpretation of results: M.D.; original manuscript draft: M.D.; manuscript review and editing: Y.L., B.G.; supervision and internal resource support: Z.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding. Z.Y. provided internal or personal resources for the work; no external grant number applies.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The source data are available from the District of Columbia, New York City, and Vancouver police open-data portals and from the Capital Bikeshare, Citi Bike, and Mobi system-data portals cited in Section 4.1 (all accessed 18 July 2026). The reproducible analysis project preserves source-file hashes, standardized schemas, reason-coded exclusions, configurations, and scripts. A versioned archive of analysis code, configurations, tests, and 24 anonymized joint-state frequency tables is publicly available at <https://doi.org/10.5281/zenodo.22936346>. The deposited tables permit independent recalculation of the primary city–domain–period entropy, CMI, and DELB point estimates. The archive is not a complete city-data reproduction package: it does not contain the inputs needed to rerun bootstrap intervals, randomization tests, multiscale or local analyses, or held-out model fitting. The public GitHub repository at <https://github.com/eveyear/delb-urban-crime-prediction> provides the revised manuscript source and analysis code, with the same data-reuse limitations as the archive. Raw and processed data are not redistributed; users must obtain them from the cited providers and comply with the providers' conditions.

Acknowledgments: The authors thank the public agencies and bicycle-system operators that made the underlying data available.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CMI	Conditional mutual information
DELB	Discrete Entropy Lower Bound
FDR	False discovery rate
GLM	Generalized linear model
KL	Kullback–Leibler
MAE	Mean absolute error
MSE	Mean squared error
RMSE	Root mean squared error

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